# Causal Inference: Identification, Estimation, and Experimental Design

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**Abstract**

This exposition reviews the identification and estimation of causal effects in randomized and observational studies. The potential-outcomes framework is used to distinguish causal estimands from observed associations and to state the assumptions underlying covariate adjustment, longitudinal comparisons, instrumental variables, synthetic controls, and regression discontinuity designs. Methods for heterogeneous treatment effects and policy evaluation are also considered. A final section addresses variance reduction, allocation diagnostics, reproducible randomization, statistical power, and sample-size determination in controlled experiments. Mathematical derivations are accompanied by illustrative implementations.

## 1 Study Design and Selection of Causal Methods

The appropriate method is determined primarily by how treatment was assigned, which variables were measured before treatment, and what source of variation can identify the causal effect. The presence of confounding cannot generally be decided from correlations or significance tests alone; it is a causal judgment based on subject-matter knowledge, study design, and a causal diagram.

### 1.1 Randomized treatment assignment

In a well-conducted randomized controlled trial (RCT), treatment assignment is independent of the potential outcomes:

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A.$$

Under consistency and no interference, the difference in observed group means identifies the ATE:

$$\tau_{\text{ATE}} = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0].$$

A difference-in-means estimator or an unadjusted regression of $Y$ on $A$ therefore estimates this identified contrast. Adding prognostic covariates can improve precision and account for chance imbalance, but it is not needed to remove systematic confounding created by treatment assignment. Noncompliance, attrition, missing outcomes, or treatment interference can still create other causal problems even when the original assignment was randomized.

### 1.2 Observational treatment with measured confounders

In observational data, treatment choice often depends on patient, clinician, or institutional characteristics. If a sufficient set of pre-treatment confounders $X$ has been measured, one may assume conditional exchangeability:

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X.$$

Together with consistency, no interference, and positivity, this identifies the ATE through covariate adjustment:

$$\tau_{\text{ATE}} = \mathbb{E}_X[\mathbb{E}[Y \mid A = 1, X] - \mathbb{E}[Y \mid A = 0, X]] .$$

Several methods implement this same identification strategy:

**Outcome regression**
: models $\mathbb{E}[Y \mid A, X]$ and is convenient when the outcome model can be specified adequately.

**Matching or propensity-score matching**
: constructs treated and control samples with similar measured baseline characteristics. It is useful for making overlap and comparability explicit, and often targets the ATT.

**Inverse probability weighting (IPW)**
: models $\Pr(A = 1 \mid X)$ and reweights observations to create a pseudo-population in which measured covariates are balanced. Extreme propensity scores can produce unstable weights.

**Doubly robust estimation**
: combines an outcome model with a propensity-score model. Estimators such as AIPW are consistent if either model is correctly specified, provided the causal identification assumptions hold.

**Double machine learning (DML)**
: uses flexible prediction methods and sample splitting to estimate nuisance functions while reducing regularization bias. It is useful with many covariates, but machine learning does not eliminate unmeasured confounding.

The adjustment set requires causal justification. Inclusion of mediators or colliders can change the estimand or introduce bias. Covariate balance and overlap are essential diagnostics for matching and weighting procedures.

### 1.3 Identification in the presence of unmeasured confounding

If important confounders are unmeasured, regression, matching, PSM, IPW, and doubly robust estimation do not automatically solve the problem. Identification must instead come from a credible design or quasi-experimental source of variation:

**Instrumental variables (IV)**
: may be used when a variable changes treatment receipt but has no direct path to the outcome and is independent of the relevant potential outcomes. With heterogeneous effects, IV generally identifies a LATE for compliers, rather than the population ATE.

**Regression discontinuity design (RDD)**
: is appropriate when treatment assignment changes at a known threshold of a running variable. It estimates a local effect near that threshold and relies on smooth potential outcomes and absence of precise manipulation.

**Difference-in-differences (DiD)**
: is appropriate when treated and comparison groups are observed before and after an intervention. It removes fixed group differences and common time shocks under the parallel-trends and no-anticipation assumptions.

**Synthetic control**
: is useful when one or a few aggregate units are treated and a donor pool is available over many

pre-treatment periods. A weighted combination of donors supplies the untreated counterfactual, provided the close pre-treatment relationship would have continued without treatment.

**Interrupted time series or BSTS**
can be considered when one outcome series has a clear intervention date and a long, stable pre-intervention history. Causal interpretation requires that no concurrent event or structural change explains the post-period deviation. Bayesian structural time series does not remove this design requirement merely by providing a flexible forecast.

These approaches permit identification without conditional exchangeability given the observed baseline covariates, provided their respective design-specific identifying assumptions hold.

### 1.4 Treatment-effect heterogeneity and method choice

If the main goal is to determine who benefits most, conditional effects

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

may be estimated using interaction models, causal forests, or uplift modeling. Uplift methods are most credible with randomized treatment or after a valid causal identification strategy has been established. Predicting outcomes accurately is not, by itself, the same as estimating treatment effects.

Method selection proceeds from specification of the estimand and treatment assignment mechanism to assessment of the identifying assumptions. The availability of measured confounders and credible sources of exogenous variation determines which identification strategies are defensible. Software availability and predictive performance alone do not establish causal validity.

## 2 Causal Framework and Identification

### 2.1 Treatment, Outcomes, and Potential Outcomes

Let $A_i \in \{0, 1\}$ denote the treatment or exposure for individual $i$, where $A_i = 1$ indicates treatment and $A_i = 0$ indicates control. Let $X_i$ denote baseline (pre-treatment) covariates and $Y_i$ the observed outcome. A treatment may represent a clinical intervention, a policy, or another well-defined exposure.

The **potential outcomes** $Y_i(1)$ and $Y_i(0)$ are the outcomes that individual $i$ would experience under treatment and control, respectively. The potential outcome under the treatment not actually received is **counterfactual**. Under consistency,

$$Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0).$$

Only one of these two potential outcomes is observed for each individual. This missing counterfactual is the **fundamental problem of causal inference**.

### 2.2 Causal Estimands

A **causal estimand** is the quantity a study aims to learn about a specified population under specified treatment strategies. It should also specify the outcome, follow-up time, and effect scale. The definitions below use the difference scale; for a binary outcome, they are risk differences.

**Individual Treatment Effect (ITE).**

The effect for individual $i$ is

$$\tau_i = Y_i(1) - Y_i(0).$$

It is generally not identifiable from observed data because the two potential outcomes cannot both be observed for the same individual.

**Average Treatment Effect (ATE).**

The average effect in the target population is

$$\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)].$$

It compares the mean outcome if everyone in that population received treatment with the mean outcome if everyone received control.

**Average Treatment Effect on the Treated (ATT).**

The average effect among individuals who actually received treatment is

$$\tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid A = 1].$$

It compares their outcome under treatment with what their outcome would have been under control.

**Average Treatment Effect on the Controls (ATC).**

Also called the average treatment effect on the untreated (ATU), this is

$$\tau_{\text{ATC}} = \mathbb{E}[Y(1) - Y(0) \mid A = 0].$$

Within the same population,

$$\tau_{\text{ATE}} = \Pr(A = 1)\tau_{\text{ATT}} + \Pr(A = 0)\tau_{\text{ATC}}.$$

**Conditional Average Treatment Effect (CATE).**

The average effect among individuals with covariates $X = x$ is

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

Variation in $\tau(x)$ across covariate values is called **treatment effect heterogeneity**. CATE is a subgroup average, not an individual treatment effect.

### 2.3 Association, Confounding, and Variable Roles

**Association** describes observed differences, such as $\mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]$. This contrast need not equal the ATE: treated and untreated individuals may differ in their underlying risk even without treatment.

A **confounder** is a common cause of treatment and outcome. For example, baseline disease severity may influence both treatment choice and subsequent recovery. **Confounding** arises when such causal structure prevents an observed treatment–outcome association from representing the causal effect. Association with both treatment and outcome alone is not sufficient to classify a variable as a confounder.

### 2.3.1 Observed difference and confounding bias

Define the crude, unadjusted difference in observed outcomes as

$$\Delta_{\text{obs}} = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0].$$

The notation $\Delta_{\text{obs}}$ denotes a cross-sectional contrast; $\Delta Y$ is reserved for a temporal outcome change. By consistency,

$$\Delta_{\text{obs}} = \mathbb{E}[Y(1) \mid A = 1] - \mathbb{E}[Y(0) \mid A = 0].$$

This compares outcomes from two different populations. To separate the causal effect from the pre-existing difference between them, add and subtract the treated population's mean outcome under no treatment:

$$\begin{aligned}
\Delta_{\text{obs}} &= \mathbb{E}[Y(1) \mid A = 1] - \mathbb{E}[Y(0) \mid A = 0] \\
&= \mathbb{E}[Y(1) \mid A = 1] - \mathbb{E}[Y(0) \mid A = 1] \\
&\quad + \mathbb{E}[Y(0) \mid A = 1] - \mathbb{E}[Y(0) \mid A = 0] \\
&= \underbrace{\mathbb{E}[Y(1) - Y(0) \mid A = 1]}_{\tau_{\text{ATT}}} + \underbrace{\{\mathbb{E}[Y(0) \mid A = 1] - \mathbb{E}[Y(0) \mid A = 0]\}}_{B_{\text{selection}}}.
\end{aligned}$$

Therefore,

$$\boxed{\Delta_{\text{obs}} = \tau_{\text{ATT}} + B_{\text{selection}}.}$$

The selection-bias term $B_{\text{selection}}$ is the difference between the groups in their mean untreated potential outcomes. It vanishes under unconditional exchangeability of the untreated potential outcome,

$$Y(0) \perp\!\!\!\perp A \quad \Longrightarrow \quad B_{\text{selection}} = 0.$$

Its sign determines the direction of bias. For example, if sicker patients are more likely to receive treatment and larger $Y$ means a worse outcome, then

$$\mathbb{E}[Y(0) \mid A = 1] > \mathbb{E}[Y(0) \mid A = 0],$$

so $B_{\text{selection}} > 0$ and the crude contrast is biased upward relative to the ATT, understating a beneficial reduction in the outcome.

This decomposition is centered on the ATT. If the target is the ATE, let $p = \Pr(A = 1)$. Because

$$\tau_{\text{ATE}} = p\tau_{\text{ATT}} + (1 - p)\tau_{\text{ATC}},$$

the bias from treating the crude difference as an ATE is

$$\Delta_{\text{obs}} - \tau_{\text{ATE}} = B_{\text{selection}} + (1 - p)(\tau_{\text{ATT}} - \tau_{\text{ATC}}).$$

Thus the crude contrast may differ from the ATE because of both selection and treatment-effect heterogeneity. If treatment effects are constant, then $\tau_{\text{ATT}} = \tau_{\text{ATC}} = \tau_{\text{ATE}}$, and only the selection-bias term remains.

The same argument can be applied within a covariate stratum $X = x$:

$$\begin{aligned}
\Delta_{\text{obs}}(x) &= \mathbb{E}[Y \mid A = 1, X = x] - \mathbb{E}[Y \mid A = 0, X = x] \\
&= \mathbb{E}[Y(1) - Y(0) \mid A = 1, X = x] + B_{\text{selection}}(x),
\end{aligned}$$

where

$$B_{\text{selection}}(x) = \mathbb{E}[Y(0) \mid A = 1, X = x] - \mathbb{E}[Y(0) \mid A = 0, X = x].$$

Under conditional exchangeability,

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X,$$

$B_{\text{selection}}(x) = 0$, and the conditional distribution of potential outcomes is identical across treatment groups. Consequently,

$$\Delta_{\text{obs}}(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x] = \tau(x).$$

Averaging these stratum-specific causal contrasts over the target population, rather than over the different covariate distributions in the observed treatment groups, gives

$$\tau_{\text{ATE}} = \mathbb{E}_X[\Delta_{\text{obs}}(X)].$$

A **mediator** lies on a causal pathway from treatment to outcome ($A \to M \to Y$). Adjusting for a mediator can block part of the total effect and does not, by itself, identify a direct effect. A **collider** is a common effect of two variables ($A \to C \leftarrow U$). Conditioning on a collider can induce an association between its causes and introduce bias. A **directed acyclic graph (DAG)** represents assumed causal relationships with directed arrows and no directed cycles; it helps distinguish these variable roles and choose an adjustment set.

### 2.4 Identification Assumptions and the Propensity Score

**Identification** is the unique determination of a causal estimand from the observed-data distribution under a specified set of assumptions. **Estimation** uses a finite sample to approximate that identified quantity. A precise estimate need not have a valid causal interpretation if the identifying assumptions fail.

For a binary treatment, the standard assumptions for identifying the ATE by baseline covariate adjustment are:

**Consistency.** If $A_i = a$, then $Y_i = Y_i(a)$. The intervention must be sufficiently well defined for the potential outcome to have a clear meaning. The **Stable Unit Treatment Value Assumption (SUTVA)** combines no interference between individuals with no hidden treatment versions: one individual's outcome does not depend on others' treatment, and the treatment label does not hide versions with different effects.

**Conditional exchangeability.** Also called no unmeasured confounding, this requires

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X.$$

Within strata of the adjustment covariates, treatment assignment is independent of the potential outcomes. This assumption is generally not testable from observed data alone.

**Positivity (overlap).** Both treatment options must be possible at covariate values occurring in the target population:

$$0 < \Pr(A = 1 \mid X = x) < 1.$$

Limited overlap makes estimation unstable; absent overlap prevents identification by adjustment without additional assumptions.

The **propensity score** is the conditional probability of receiving treatment given baseline covariates,

$$e(x) = \Pr(A = 1 \mid X = x).$$

It is used in propensity score matching and inverse probability weighting to balance measured baseline covariates. These methods do not automatically remove unmeasured confounding.

Under consistency, no interference, conditional exchangeability, and positivity, the ATE is identified by the adjustment formula

$$\begin{aligned}
\tau_{\text{ATE}} &= \mathbb{E}[Y(1) - Y(0)] \\
&= \mathbb{E}_X[\mathbb{E}[Y(1) - Y(0) \mid X]] \\
&= \mathbb{E}_X[\mathbb{E}[Y(1) \mid X] - \mathbb{E}[Y(0) \mid X]] \\
&= \mathbb{E}_X[\mathbb{E}[Y(1) \mid A = 1, X] - \mathbb{E}[Y(0) \mid A = 0, X]] \\
&= \mathbb{E}_X[\mathbb{E}[Y \mid A = 1, X] - \mathbb{E}[Y \mid A = 0, X]] .
\end{aligned}$$

The derivation applies the law of iterated expectations and linearity of conditional expectation. Conditional exchangeability then permits conditioning on the corresponding treatment group: conditional on $X$, the mean potential outcome under $a$ is the same in the group with $A = a$ as in the full population. The final line follows from consistency, which gives $Y = Y(a)$ for people whose observed treatment is $A = a$. No interference ensures that $Y_i(1)$ and $Y_i(0)$ are well-defined as potential outcomes for individual $i$, independent of other individuals' treatment assignments. Positivity ensures that, for each relevant value of $X$, both conditional means in the final line are supported by observed data; it does not correspond to an algebraic equality in the derivation. The outer expectation averages over the target population's covariate distribution. Other designs, such as instrumental variables or difference-in-differences, rely on different identifying assumptions and may target different causal estimands.

# 3 Methods for Causal Effect Estimation

This section develops covariate-adjustment estimators, longitudinal and instrument-based identification strategies, and methods for heterogeneous treatment effects. Each method is presented in terms of its estimand, identifying assumptions, estimation procedure, and inferential limitations. The code examples illustrate implementation and do not replace assessment of the study design and identifying assumptions.

## 3.1 Regression

Regression adjustment estimates a causal effect by modeling the conditional mean outcome given treatment and pre-treatment covariates. Define

$$m(a, x) = \mathbb{E}[Y \mid A = a, X = x].$$

Under consistency, conditional exchangeability, and positivity,

$$\begin{aligned}
\mathbb{E}[Y(a)] &= \mathbb{E}_X[\mathbb{E}[Y(a) \mid X]] && \text{(law of iterated expectations)} \\
&= \mathbb{E}_X[\mathbb{E}[Y(a) \mid A = a, X]] && \text{(conditional exchangeability)} \\
&= \mathbb{E}_X[\mathbb{E}[Y \mid A = a, X]] && \text{(consistency)} \\
&= \mathbb{E}_X[m(a, X)].
\end{aligned}$$

Consequently, the ATE is identified by the **g-formula**

$$\tau_{\mathrm{ATE}} = \mathbb{E}_X[m(1, X) - m(0, X)].$$

Regression supplies an estimate $\widehat{m}(a, x)$ of the unknown conditional mean; averaging its two treatment predictions over the observed covariate distribution gives the regression-standardized estimator

$$\widehat{\tau}_{\mathrm{gcomp}} = \frac{1}{n} \sum_{i=1}^{n} \left\{ \widehat{m}(1, X_i) - \widehat{m}(0, X_i) \right\}.$$

Thus, regression is an estimation device. Its causal interpretation still comes from the identification assumptions and correct adjustment-set choice.

### 3.1.1 Linear outcome regression

Suppose that the conditional mean is linear and the treatment effect is constant:

$$Y_i = \beta_0 + \tau A_i + X_i^{\mathsf{T}} \beta + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid A_i, X_i] = 0.$$

Then

$$\begin{aligned} m(1, x) &= \beta_0 + \tau + x^{\mathsf{T}} \beta, \\ m(0, x) &= \beta_0 + x^{\mathsf{T}} \beta, \end{aligned}$$

so that

$$m(1, x) - m(0, x) = \tau \quad \Longrightarrow \quad \tau_{\mathrm{ATE}} = \tau.$$

Therefore, under the causal assumptions and correct model specification, the coefficient of $A$ is the conditional effect and, because the effect is constant, also the marginal ATE on the difference scale.

To derive the ordinary least-squares (OLS) estimator, let $Z_i = (1, A_i, X_i^{\mathsf{T}})^{\mathsf{T}}$, stack the observations into the design matrix $Z$, and write

$$Y = Z\theta + \varepsilon, \qquad \theta = (\beta_0, \tau, \beta^{\mathsf{T}})^{\mathsf{T}}.$$

OLS minimizes the residual sum of squares

$$Q(\theta) = (Y - Z\theta)^{\mathsf{T}} (Y - Z\theta).$$

Differentiating and setting the gradient to zero gives

$$\begin{aligned} \frac{\partial Q(\theta)}{\partial \theta} &= -2Z^{\mathsf{T}}(Y - Z\theta) = 0, \\ Z^{\mathsf{T}} Z \widehat{\theta} &= Z^{\mathsf{T}} Y, \\ \widehat{\theta} &= (Z^{\mathsf{T}} Z)^{-1} Z^{\mathsf{T}} Y, \end{aligned}$$

provided $Z^{\mathsf{T}} Z$ is nonsingular. The component corresponding to $A$ is $\widehat{\tau}$.

The Frisch–Waugh–Lovell theorem clarifies what this coefficient compares. Let $W = [\mathbf{1}, X]$ and

$$M_W = I - W(W^{\mathsf{T}} W)^{-1} W^{\mathsf{T}}.$$

The vectors $\widetilde{A} = M_W A$ and $\widetilde{Y} = M_W Y$ are treatment and outcome residuals after their linear projections on $X$. Then

$$\widehat{\tau} = \frac{A^\mathsf{T} M_W Y}{A^\mathsf{T} M_W A} = \frac{\sum_i \widetilde{A}_i \widetilde{Y}_i}{\sum_i \widetilde{A}_i^2}.$$

Hence adjusted regression compares outcome variation with treatment variation remaining among observations with similar covariate profiles. If treatment is nearly determined by $X$, little residual treatment variation remains; this is the regression manifestation of poor positivity and leads to large variance and extrapolation.

### 3.1.2 Confounding and omitted-variable bias

Consider one confounder $X$ and the data-generating model

$$Y = \beta_0 + \tau A + \beta_X X + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid A, X] = 0.$$

If $X$ is omitted and $Y$ is regressed only on $A$, the population slope is

$$\begin{aligned}
\tau_{\text{unadjusted}} &= \frac{\operatorname{Cov}(A, Y)}{\operatorname{Var}(A)} \\
&= \frac{\operatorname{Cov}(A, \beta_0 + \tau A + \beta_X X + \varepsilon)}{\operatorname{Var}(A)} \\
&= \tau + \beta_X \frac{\operatorname{Cov}(A, X)}{\operatorname{Var}(A)}.
\end{aligned}$$

The second term is the omitted-variable bias. It is nonzero when $X$ predicts the outcome ($\beta_X \neq 0$) and is associated with treatment ($\operatorname{Cov}(A, X) \neq 0$). Including an appropriate pre-treatment confounder can remove this bias under correct specification, whereas adjusting for a mediator or collider can change the estimand or introduce bias.

### 3.1.3 Effect heterogeneity and nonlinear specifications

If the treatment effect varies with $X$, include treatment–covariate interactions. For a scalar covariate,

$$\mathbb{E}[Y \mid A, X] = \beta_0 + \tau A + \beta_X X + \gamma A X.$$

The conditional effect is

$$m(1, x) - m(0, x) = \tau + \gamma x,$$

and standardization gives

$$\tau_{\text{ATE}} = \tau + \gamma \mathbb{E}[X].$$

Thus $\tau$ alone is the effect at $X = 0$, not generally the ATE. Centering $X$ at its sample mean makes the treatment coefficient easier to interpret, but prediction and standardization remain the general procedure.

Flexible terms such as splines, polynomials, and interactions may be needed to estimate $m(a, x)$ adequately. For binary outcomes, logistic regression models

$$\operatorname{logit} \Pr(Y = 1 \mid A, X) = \beta_0 + \tau A + X^\mathsf{T} \beta.$$

Here $\exp(\tau)$ is a conditional odds ratio, not an ATE or a risk ratio. To estimate the ATE on the risk-difference scale, predict each individual's risk twice and average the differences:

$$\widehat{\tau}_{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{p}_i(1) - \widehat{p}_i(0)\right\}.$$

Model diagnostics should assess functional form, influential observations, and treatment overlap. Heteroskedasticity-robust standard errors protect inference against nonconstant error variance, but they do not repair confounding, poor overlap, measurement error, or outcome-model misspecification. Bootstrap confidence intervals can incorporate the full standardization procedure.

### 3.1.4 Python implementation

The following example uses `statsmodels`. The columns are a continuous outcome `Y`, binary treatment `A`, and pre-treatment covariates `age` and `severity`. The first fit reports the constant-effect coefficient with an HC3 robust standard error. The second calculation uses the fitted model as a g-computation estimator, which also works when treatment interactions are added to the formula.

```
import numpy as np
import statsmodels.formula.api as smf

# Adjust only for justified pre-treatment covariates.
formula = "Y ~ A + age + severity"
model = smf.ols(formula, data=df).fit(cov_type="HC3")
print(model.summary())
print("Treatment coefficient:", model.params["A"])

# Regression standardization (g-computation) for the ATE.
treated = df.assign(A=1)
control = df.assign(A=0)
pred_treated = model.predict(treated)
pred_control = model.predict(control)
individual_contrasts = pred_treated - pred_control
ate = individual_contrasts.mean()

# Bootstrap CI for the complete procedure.
rng = np.random.default_rng(2026)
boot = []
for _ in range(1000):
    sample = df.iloc[rng.integers(0, len(df), len(df))]
    fit_b = smf.ols("Y ~ A + age + severity", data=sample).fit()
    boot.append((fit_b.predict(sample.assign(A=1))
                 - fit_b.predict(sample.assign(A=0))).mean())

ci_low, ci_high = np.quantile(boot, [0.025, 0.975])
print("Standardized ATE:", ate)
print("95% bootstrap CI:", (ci_low, ci_high))
```

## 3.2 Matching

Matching constructs a comparison group before estimating the treatment effect. For each unit, it seeks units receiving the other treatment whose pre-treatment covariates are similar. For a treated

person with $X_i = x$, the outcomes of comparable controls estimate the missing counterfactual $Y_i(0)$.

The causal justification is conditional exchangeability,

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X,$$

combined with consistency and positivity. In particular,

$$\mathbb{E}[Y(0) \mid A = 1, X = x] = \mathbb{E}[Y(0) \mid A = 0, X = x] = \mathbb{E}[Y \mid A = 0, X = x].$$

Thus a control with covariates close to $x$ provides information about a treated person's unobserved control potential outcome. Matching balances only the measured covariates used to define similarity; it cannot repair unmeasured confounding.

### 3.2.1 Nearest-neighbor matching and the ATT

Let $\mathcal{T} = \{i : A_i = 1\}$ and $\mathcal{C} = \{j : A_j = 0\}$ be the treated and control indices, and let $n_1 = |\mathcal{T}|$. Let $d(X_i, X_j)$ measure the distance between covariate vectors. A common choice after standardizing continuous covariates is the Mahalanobis distance,

$$d_M(X_i, X_j) = \sqrt{(X_i - X_j)^\intercal S_X^{-1}(X_i - X_j)},$$

where $S_X$ is the covariate sample covariance matrix. Exact matching can be used for important discrete covariates by allowing matches only within the same category.

For nearest-neighbor matching with replacement, define the matched control for treated unit $i$ as

$$j(i) = \arg\min_{j \in \mathcal{C}} d(X_i, X_j).$$

The same control may be used more than once. The matching estimator is

$$\widehat{\tau}_{\text{ATT}}^{\text{match}} = \frac{1}{n_1} \sum_{i \in \mathcal{T}} \left\{ Y_i - Y_{j(i)} \right\}.$$

The observed $Y_i = Y_i(1)$ is paired with $Y_{j(i)}$, which estimates the missing $Y_i(0)$. It targets the ATT because the average is taken over the treated population.

With $K$ matches per treated unit, let $\mathcal{M}(i)$ be the set of matches and $K_i = |\mathcal{M}(i)|$. Then

$$\widehat{\tau}_{\text{ATT}}^{\text{match}} = \frac{1}{n_1} \sum_{i \in \mathcal{T}} \left\{ Y_i - \frac{1}{K_i} \sum_{j \in \mathcal{M}(i)} Y_j \right\}.$$

More matches can reduce variance but can increase bias if they are distant. Matching with replacement often reduces distance and bias, though it may give large weight to a few controls.

To target the ATE, both potential outcomes must be imputed for every unit. One symmetric nearest-neighbor estimator is

$$\widehat{\tau}_{\text{ATE}}^{\text{match}} = \frac{1}{n} \left[ \sum_{i \in \mathcal{T}} \{Y_i - Y_{j(i)}\} + \sum_{i \in \mathcal{C}} \{Y_{k(i)} - Y_i\} \right],$$

where $j(i)$ is a matched control for treated $i$, and $k(i)$ is a matched treated unit for control $i$. The matched target population should always be reported, particularly when unmatched units are discarded.

### 3.2.2 Overlap, calipers, and diagnostics

Positivity requires support for both treatments at relevant covariate values. In finite samples, reliable matching additionally requires sufficiently close opposite-treatment observations. Large matching distances indicate limited comparability and increased dependence on counterfactual extrapolation. A **caliper** discards proposed matches whose distance exceeds a chosen threshold. When matching on the propensity score $e(X)$, a common starting rule is a caliper of 0.2 standard deviations of $\text{logit}\{e(X)\}$; it should be justified and reported rather than treated as universal.

Matching quality should be assessed without inspecting outcomes. For a continuous covariate $X_r$, the standardized mean difference (SMD) is

$$\text{SMD}(X_r) = \frac{\overline{X}_{r,1} - \overline{X}_{r,0}}{\sqrt{(s^2_{r,1} + s^2_{r,0})/2}}.$$

After matching, compute the control mean and variance with matching weights, including repeated controls. Values of $|\,\text{SMD}\,| < 0.1$ are a common diagnostic, not proof of exchangeability. Also inspect covariate variances, important nonlinear terms and interactions, propensity-score distributions, matching distances, and discarded observations.

### 3.2.3 Python implementation

The following example implements one-to-one nearest-neighbor matching with replacement on standardized covariates to estimate the ATT. Applied analyses require a causally justified covariate set, overlap assessment, and post-matching balance diagnostics.

```
import numpy as np
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import StandardScaler

covariates = ["age", "severity", "baseline_biomarker"]
treated = df.loc[df["A"] == 1].copy()
control = df.loc[df["A"] == 0].copy()

# Make Euclidean distance comparable across covariates.
scaler = StandardScaler().fit(df[covariates])
x_treated = scaler.transform(treated[covariates])
x_control = scaler.transform(control[covariates])

# One nearest control per treated person; controls may be reused.
nn = NearestNeighbors(n_neighbors=1).fit(x_control)
distances, indices = nn.kneighbors(x_treated)
matched_control = control.iloc[indices[:, 0]].copy()

pair_effects = (treated["Y"].to_numpy()
                - matched_control["Y"].to_numpy())
att = pair_effects.mean()
print("Matched ATT:", att)
print("Largest matching distance:", distances.max())
```

### 3.3 PSM/IPW

Propensity-score matching (PSM) and inverse probability weighting (IPW) both use the propensity score

$$e(x) := \Pr(A = 1 \mid X = x),$$

but they use it in different ways. PSM uses $e(X)$ as a one-dimensional distance for finding comparable units. IPW uses $e(X)$ to assign observation weights and construct a pseudo-population. The slash in "PSM/IPW" does not mean that they are the same estimator.

Both methods rely on the same causal identification assumptions:

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X, \qquad 0 < e(X) < 1,$$

together with consistency and no interference. The first condition says that $X$ contains a sufficient set of measured pre-treatment confounders; neither PSM nor IPW removes bias from an important confounder omitted from $X$. The second condition requires treated and control observations at relevant covariate values.

The true propensity score is a *balancing score*:

$$A \perp\!\!\!\perp X \mid e(X).$$

This property follows from the definition of $e(X)$ and does not require conditional exchangeability. Indeed, the law of iterated expectations gives

$$\Pr(A = 1 \mid e(X)) = \mathbb{E}[\Pr(A = 1 \mid X) \mid e(X)] = e(X) = \Pr(A = 1 \mid X, e(X)).$$

Because $A$ is binary, equality of these probabilities establishes the conditional independence. If conditional exchangeability given $X$ also holds, then, under positivity,

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid e(X).$$

Thus units with the same true propensity score have the same distribution of measured baseline covariates across treatment groups. This dimension reduction does not mean that the variables in $X$ may be selected only by their ability to predict treatment. They must first be selected using causal knowledge.

In practice, $e(X)$ is unknown and is estimated, often with logistic regression,

$$\operatorname{logit}\{e(X_i)\} = \alpha_0 + \alpha^\mathsf{T} X_i, \qquad \widehat{e}_i = \operatorname{expit}(\widehat{\alpha}_0 + \widehat{\alpha}^\mathsf{T} X_i),$$

or with a flexible prediction method. A high treatment-prediction accuracy is not the goal by itself; the relevant empirical goals are overlap and balance of the confounders after matching or weighting.

#### 3.3.1 Propensity-score matching

Let $\mathcal{T} = \{i : A_i = 1\}$ and $\mathcal{C} = \{j : A_j = 0\}$. In one-to-one nearest-neighbor PSM for the ATT, a treated unit $i$ is paired with

$$j(i) \in \arg\min_{j \in \mathcal{C}} \left|\operatorname{logit}(\widehat{e}_i) - \operatorname{logit}(\widehat{e}_j)\right|.$$

Matching is commonly performed on the logit propensity score because distances near zero and one are compressed on the probability scale. The matched ATT is

$$\widehat{\tau}_{\mathrm{ATT}}^{\mathrm{PSM}} = \frac{1}{n_1} \sum_{i \in \mathcal{T}} \{Y_i - Y_{j(i)}\}, \qquad n_1 = |\mathcal{T}|.$$

A caliper may reject a pair when its distance exceeds a prespecified threshold. Discarding treated units changes the population represented by the estimate, so the number and characteristics of discarded units must be reported. Matching with replacement can improve match quality but requires accounting for the fact that the same control may represent several treated units.

PSM does not guarantee balance merely because the propensity-score model converged. After matching, compare every important covariate between groups using standardized mean differences, distributions, variances, and relevant interactions. If balance is poor, revise the propensity specification or the matching design without using the outcome to search for a favorable result.

### 3.3.2 Inverse probability weighting

IPW gives greater weight to observations receiving a treatment that was unlikely given their covariates. For the ATE, the Horvitz–Thompson weights are

$$\omega_i^{\mathrm{ATE}} = \frac{A_i}{e(X_i)} + \frac{1 - A_i}{1 - e(X_i)}.$$

For example, a treated unit with $e(X_i) = 0.2$ receives weight $1/0.2 = 5$, because this type of unit is uncommon in the treated group. A treated unit with $e(X_i) = 0.8$ receives weight $1/0.8 = 1.25$.

The identifying equality for the treated potential-outcome mean is

$$\begin{aligned} \mathbb{E}\left[\frac{AY}{e(X)}\right] &= \mathbb{E}_X\left[\frac{\mathbb{E}[AY \mid X]}{e(X)}\right] \\ &= \mathbb{E}_X\left[\frac{e(X)\mathbb{E}[Y \mid A = 1, X]}{e(X)}\right] \\ &= \mathbb{E}_X[\mathbb{E}\{Y(1) \mid X\}] = \mathbb{E}[Y(1)]. \end{aligned}$$

The second line uses the definition of $e(X)$; the third uses consistency and conditional exchangeability. Similarly,

$$\mathbb{E}\left[\frac{(1 - A)Y}{1 - e(X)}\right] = \mathbb{E}[Y(0)].$$

Therefore the population IPW representation of the ATE is

$$\tau_{\mathrm{ATE}} = \mathbb{E}\left[\frac{AY}{e(X)} - \frac{(1 - A)Y}{1 - e(X)}\right].$$

Replacing $e(X_i)$ with $\widehat{e}_i$ gives the Horvitz–Thompson estimator

$$\widehat{\tau}_{\mathrm{ATE}}^{\mathrm{HT}} = \frac{1}{n} \sum_{i=1}^{n} \left\{\frac{A_i Y_i}{\widehat{e}_i} - \frac{(1 - A_i)Y_i}{1 - \widehat{e}_i}\right\}.$$

A commonly used normalized, or Hájek, version estimates the two potential outcome means separately:

$$\widehat{\tau}_{\mathrm{ATE}}^{\mathrm{Hajek}} = \frac{\sum_i A_i Y_i/\widehat{e}_i}{\sum_i A_i/\widehat{e}_i} - \frac{\sum_i (1-A_i) Y_i/(1-\widehat{e}_i)}{\sum_i (1-A_i)/(1-\widehat{e}_i)}.$$

Normalization often improves finite-sample stability. Stabilized ATE weights replace the numerators 1 by the marginal treatment probabilities:

$$\omega_i^{\mathrm{stab}} = \frac{A_i \Pr(A=1)}{e(X_i)} + \frac{(1-A_i)\Pr(A=0)}{1-e(X_i)}.$$

IPW can also target the ATT. Treated units receive weight one, while controls are reweighted to resemble the treated covariate distribution:

$$\omega_i^{\mathrm{ATT}} = A_i + (1-A_i)\frac{e(X_i)}{1-e(X_i)}.$$

Thus the estimand must be chosen before selecting weights: ATE and ATT weights answer different questions.

### 3.3.3 Overlap, diagnostics, and uncertainty

When $\widehat{e}_i$ is close to zero for a treated unit or close to one for a control unit, inverse weights become extreme. This is not merely a numerical problem: it indicates that the data contain little information about the missing treatment condition for those covariate profiles. Inspect propensity score distributions by treatment, effective sample size,

$$n_{\mathrm{eff}} := \frac{(\sum_i \omega_i)^2}{\sum_i \omega_i^2},$$

the largest weights, and weighted covariate balance. Trimming, truncation, or overlap weights can reduce variance, but they may change the target population and should be specified and reported explicitly.

Uncertainty calculations must account for estimation of the propensity score and for the matching or weighting procedure. Ordinary unweighted standard errors applied after constructing weights are generally inappropriate. Use an estimator-specific sandwich variance, a justified bootstrap of the complete procedure, or a package implementing the required influence-function variance.

### 3.3.4 Python implementation

The following example estimates the propensity score once, uses it for one-to-one PSM targeting the ATT, and then uses Hájek IPW targeting the ATE. The implementation illustrates point estimation. Applied analyses additionally require balance tables, overlap plots, prespecified caliper or trimming rules where applicable, and estimator-appropriate uncertainty quantification.

```
import numpy as np
from scipy.special import logit
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import NearestNeighbors

covariates = ["age", "severity", "baseline_biomarker"]
x = df[covariates]
```

```
a = df["A"].to_numpy()
y = df["Y"].to_numpy()

# Model treatment using pre-treatment confounders.
ps_model = LogisticRegression(max_iter=2000)
ps_model.fit(x, a)
ps = ps_model.predict_proba(x)[:, 1]
ps = np.clip(ps, 1e-6, 1 - 1e-6)

# PSM: nearest control on logit propensity; targets ATT.
treated_idx = np.flatnonzero(a == 1)
control_idx = np.flatnonzero(a == 0)
logit_ps = logit(ps).reshape(-1, 1)

nn = NearestNeighbors(n_neighbors=1)
nn.fit(logit_ps[control_idx])
distance, neighbor = nn.kneighbors(logit_ps[treated_idx])
matched_control_idx = control_idx[neighbor[:, 0]]
att_psm = np.mean(y[treated_idx] - y[matched_control_idx])

# IPW: normalized (Hajek) weighted means; targets ATE.
w1 = a / ps
w0 = (1 - a) / (1 - ps)
mu1 = np.sum(w1 * y) / np.sum(w1)
mu0 = np.sum(w0 * y) / np.sum(w0)
ate_ipw = mu1 - mu0

print("PSM ATT:", att_psm)
print("IPW ATE:", ate_ipw)
print("Largest matching distance:", distance.max())
print("Largest ATE weight:", np.maximum(w1, w0).max())
```

### 3.4 Difference-in-Differences (DiD)

Difference-in-differences estimates an intervention effect by comparing the outcome change over time in a treated group with the contemporaneous change in an untreated comparison group. The second difference subtracts the background time trend that the treated group would have experienced without treatment.

Consider two groups, $A_i \in \{0,1\}$, and two periods, $t \in \{0,1\}$. Group $A_i = 1$ begins treatment in the post-treatment period $t = 1$; group $A_i = 0$ is untreated in both periods. Here $A_i$ denotes membership in the group that receives treatment at $t = 1$, not treatment receipt at every period. Actual period-specific treatment status is therefore

$$D_{it} = A_i \times \text{Post}_t, \qquad \text{Post}_t = \mathbf{1}\{t = 1\}.$$

Let

$$\mu_{at} = \mathbb{E}[Y_{it} \mid A_i = a].$$

The two-period DiD estimand is

$$\begin{aligned} \tau_{\text{DiD}} &= (\mu_{11} - \mu_{10}) - (\mu_{01} - \mu_{00}) \\ &= \underbrace{(\text{treated post} - \text{treated pre})}_{\text{observed change in the treated group}} - \underbrace{(\text{control post} - \text{control pre})}_{\text{background change}}. \end{aligned}$$

### 3.4.1 Identification by parallel trends

Let $Y_{it}(1)$ and $Y_{it}(0)$ denote the outcomes at time $t$ under treatment and no treatment. With no anticipation and no treatment at $t = 0$, the observed baseline outcome is $Y_{i0} = Y_{i0}(0)$. At $t = 1$, treated units reveal $Y_{i1}(1)$ and controls reveal $Y_{i1}(0)$. The target is the post-treatment ATT,

$$\tau_{\text{ATT},1} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid A_i = 1].$$

The missing term is the treated group's post-treatment counterfactual $\mathbb{E}[Y_{i1}(0) \mid A_i = 1]$.

The central assumption is **parallel trends** in untreated potential outcomes:

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid A_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid A_i = 0].$$

It does not require the groups to have the same baseline outcome. It requires their average changes, had neither group been treated, to be the same. Using this assumption,

$$\begin{aligned}
\tau_{\text{ATT},1} &= \mathbb{E}[Y_{i1}(1) \mid A_i = 1] - \mathbb{E}[Y_{i1}(0) \mid A_i = 1] \\
&= \mathbb{E}[Y_{i1} \mid A_i = 1] - \mathbb{E}[Y_{i0} \mid A_i = 1] \\
&\quad - \{\mathbb{E}[Y_{i1} \mid A_i = 0] - \mathbb{E}[Y_{i0} \mid A_i = 0]\} \\
&= \tau_{\text{DiD}}.
\end{aligned}$$

The second equality adds and subtracts the treated group's untreated baseline mean, applies parallel trends to the untreated change, and uses consistency and no anticipation to express the result in observed outcomes.

Identification also requires no anticipation: treatment at $t = 1$ must not affect the outcome measured at $t = 0$. Composition of the groups and outcome measurement should remain comparable over time, and no other intervention should differentially affect the treated group at the same date. Conditional parallel trends may instead be assumed given pre-treatment covariates $X$, but post-treatment covariates should not be controlled as ordinary confounders.

### 3.4.2 Regression representation and diagnostics

The canonical regression is

$$Y_{it} = \alpha + \gamma A_i + \lambda \,\text{Post}_t + \tau(A_i \times \text{Post}_t) + \varepsilon_{it}.$$

Its four fitted cell means are

$$\begin{aligned}
\widehat{\mu}_{00} &= \widehat{\alpha}, & \widehat{\mu}_{01} &= \widehat{\alpha} + \widehat{\lambda}, \\
\widehat{\mu}_{10} &= \widehat{\alpha} + \widehat{\gamma}, & \widehat{\mu}_{11} &= \widehat{\alpha} + \widehat{\gamma} + \widehat{\lambda} + \widehat{\tau}.
\end{aligned}$$

Therefore,

$$(\widehat{\mu}_{11} - \widehat{\mu}_{10}) - (\widehat{\mu}_{01} - \widehat{\mu}_{00}) = \widehat{\tau}.$$

With multiple periods, unit and time fixed effects give

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}.$$

Standard errors should normally be clustered at the level at which treatment is assigned, because observations within a unit, hospital, state, or other treatment cluster are correlated over time.

Pre-treatment trends should be plotted and examined, but failure to reject a pre-trend test does not prove parallel trends. When groups adopt treatment at different dates and effects vary across cohorts or time, the coefficient from a simple two-way fixed-effects regression can combine inappropriate comparisons. Cohort-and-time-specific modern DiD estimators are preferable in that setting.

### 3.4.3 Python implementation

For a two-group, two-period design, the interaction coefficient below equals the sample DiD. The data contain one row per unit and period.

```
import statsmodels.formula.api as smf

# A marks treated-group membership; post marks the later period.
did = smf.ols("Y ~ A * post", data=df).fit(
    cov_type="cluster",
    cov_kwds={"groups": df["unit_id"]},
)
print("DiD estimate:", did.params["A:post"])
print(did.conf_int().loc["A:post"])
```

## 3.5 Instrumental Variables (IV)

Instrumental variables identify a treatment effect when treatment is confounded but there is a variable $Z$ that changes treatment receipt without otherwise changing the outcome. Examples may include randomized treatment assignment with noncompliance, or a valid rule that shifts access to treatment. Validity comes from the causal argument for the instrument, not from a large first-stage coefficient alone.

Let $Z_i \in \{0, 1\}$ be an instrument, $A_i(z)$ the treatment individual $i$ would receive under instrument value $z$, and $Y_i(a, z)$ the corresponding potential outcome. The standard binary-IV assumptions are:

**Relevance.** The instrument changes the probability of treatment:

$$\Pr(A = 1 \mid Z = 1) \neq \Pr(A = 1 \mid Z = 0).$$

**Independence.** The instrument is as-if randomly assigned, possibly conditional on pre-treatment covariates $X$:

$$Z \perp\!\!\!\perp \{A(0), A(1), Y(0), Y(1)\} \mid X.$$

**Exclusion restriction.** The instrument affects the outcome only through the received treatment:

$$Y_i(a, z) = Y_i(a).$$

**Monotonicity.** The instrument does not push anyone in the direction opposite to its intended effect on treatment:

$$A_i(1) \geq A_i(0) \quad \text{for every } i.$$

This rules out *defiers* but allows always-takers, never-takers, and compliers.

SUTVA and a well-defined instrument are also required. Independence and the exclusion restriction are generally not testable using the observed data alone.

### 3.5.1 The Wald estimand and LATE

For a binary instrument and treatment, the reduced-form or intention-to-treat effect on the outcome is

$$\Delta_Y = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0],$$

and the first-stage effect on treatment is

$$\Delta_A = \mathbb{E}[A \mid Z = 1] - \mathbb{E}[A \mid Z = 0].$$

Under independence and exclusion,

$$\begin{aligned}\Delta_Y &= \mathbb{E}[Y_i\{A_i(1)\} - Y_i\{A_i(0)\}] \\ &= \mathbb{E}[\{Y_i(1) - Y_i(0)\}\{A_i(1) - A_i(0)\}]\,.\end{aligned}$$

Under monotonicity, $A_i(1) - A_i(0)$ equals one for compliers and zero for always-takers and never-takers. Therefore,

$$\Delta_Y = \Pr(\text{complier})\mathbb{E}[Y(1) - Y(0) \mid \text{complier}],$$

while

$$\Delta_A = \Pr(\text{complier}).$$

Dividing gives the Wald estimand

$$\tau_{\text{LATE}} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[A \mid Z = 1] - \mathbb{E}[A \mid Z = 0]} = \mathbb{E}[Y(1) - Y(0) \mid \text{complier}].$$

Thus IV generally identifies the **local average treatment effect (LATE)** among compliers whose treatment changes because of the instrument, not the population ATE. A weak instrument makes $\Delta_A$ close to zero, so the ratio becomes unstable and conventional confidence intervals can be misleading.

### 3.5.2 Two-stage least squares

For a linear model with covariates $X$, two-stage least squares (2SLS) first predicts treatment from the instrument and exogenous covariates,

$$A_i = \pi_0 + \pi_1 Z_i + X_i^{\mathsf{T}}\pi_X + v_i,$$

and then uses the instrument-generated part of treatment to estimate the outcome equation,

$$Y_i = \beta_0 + \tau A_i + X_i^{\mathsf{T}}\beta_X + u_i.$$

Let $Q = [\mathbf{1}, Z, X]$ contain the instruments and exogenous regressors. Let $R = [\mathbf{1}, A, X]$ be the outcome-equation design matrix, and define

$$P_Q = Q(Q^{\mathsf{T}}Q)^{-1}Q^{\mathsf{T}}.$$

Then the 2SLS estimator is

$$\widehat{\beta}_{\text{2SLS}} = (R^{\mathsf{T}}P_Q R)^{-1}R^{\mathsf{T}}P_Q Y.$$

Conventional OLS standard errors from a regression on fitted treatment are not valid for 2SLS. Inference requires the appropriate IV variance estimator, with heteroskedasticity or clustering addressed as warranted by the design. Reporting includes first-stage strength, substantive justification of the instrument, and the population represented by the identified effect.

### 3.5.3 Python implementation

The following uses `linearmodels`. In the formula, `A` is the endogenous treatment, `Z` its excluded instrument, and the baseline covariates enter both stages as exogenous regressors.

```
from linearmodels.iv import IV2SLS

formula = "Y ~ 1 + age + severity + [A ~ Z]"
iv = IV2SLS.from_formula(formula, data=df).fit(
    cov_type="robust"
)
print(iv.summary)
print("IV treatment effect:", iv.params["A"])
print(iv.first_stage)
```

## 3.6 Synthetic Control Method

Synthetic control is designed for settings in which one or a small number of aggregate units receive an intervention at a known time. A unit may be a country, city, hospital, school, or company. Instead of selecting a single control unit, the method constructs a weighted average of untreated units that resembles the treated unit before treatment.

Suppose unit $i = 1$ is treated after period $T_0$, while units $i = 2, \dots, J+1$ remain untreated throughout the study. Let $Y_{it}(1)$ and $Y_{it}(0)$ denote potential outcomes. The observed outcome for the treated unit is

$$Y_{1t} = \begin{cases} Y_{1t}(0), & t \le T_0, \\ Y_{1t}(1), & t > T_0. \end{cases}$$

The post-treatment effect for the treated unit at time $t$ is

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0.$$

The missing quantity is $Y_{1t}(0)$: what would have happened to the treated unit after $T_0$ without the intervention.

### 3.6.1 Constructing the synthetic control

Index the $J$ donor units by $r = 1, \dots, J$, where donor $r$ is the original unit $i = r + 1$. Define the donor-weight vector and the unit simplex by

$$\mathbf{w} = (w_1, \dots, w_J)^\mathsf{T} \in \mathbb{R}^J, \qquad \Delta_J := \left\{ \mathbf{w} \in \mathbb{R}^J : \mathbf{w} \succeq \mathbf{0}_J,\ \mathbf{1}_J^\mathsf{T}\mathbf{w} = 1 \right\}.$$

Here $\mathbf{w} \succeq \mathbf{0}_J$ means componentwise nonnegativity. Consequently, every feasible synthetic control is a convex combination of the donor units and lies in their convex hull.

Let $K$ be the number of pre-treatment predictors. Write the treated unit's predictor vector as $\mathbf{x}_1 \in \mathbb{R}^K$, and write donor $r$'s vector as $\mathbf{x}_{0r} \in \mathbb{R}^K$. Collect the donor vectors as columns of

$$\mathbf{X}_0 := [\mathbf{x}_{01}, \dots, \mathbf{x}_{0J}] \in \mathbb{R}^{K \times J}.$$

Thus $\mathbf{X}_0\mathbf{w} \in \mathbb{R}^K$ is the predictor vector of the synthetic control. For a prespecified symmetric positive semidefinite matrix $\mathbf{V} \in \mathbb{R}^{K \times K}$, define

$$\|\mathbf{a}\|_\mathbf{V}^2 := \mathbf{a}^\mathsf{T}\mathbf{V}\mathbf{a}, \qquad \mathbf{a} \in \mathbb{R}^K.$$

The set of optimal weights is

$$\mathcal{W}^* := \arg\min_{\mathbf{w}\in\Delta_J} \|\mathbf{x}_1 - \mathbf{X}_0\mathbf{w}\|_{\mathbf{V}}^2 .$$

Choose any $\widehat{\mathbf{w}} \in \mathcal{W}^*$. Writing an *argmin set* rather than a single vector allows for nonunique minimizers. The matrix $\mathbf{V}$ determines the relative importance and scaling of the predictors; it must be selected using pre-treatment information only.

The common special case using the complete pre-treatment outcome path can be written entirely in matrix form. Define

$$\mathbf{y}_1^{\text{pre}} := (Y_{11}, \ldots, Y_{1T_0})^{\mathsf{T}} \in \mathbb{R}^{T_0}$$

and

$$\mathbf{Y}_0^{\text{pre}} := \begin{bmatrix} Y_{2,1} & \cdots & Y_{J+1,1} \\ \vdots & \ddots & \vdots \\ Y_{2,T_0} & \cdots & Y_{J+1,T_0} \end{bmatrix} \in \mathbb{R}^{T_0\times J}.$$

Column $r$ of $\mathbf{Y}_0^{\text{pre}}$ is therefore the pre-treatment outcome path of donor unit $r+1$. With equal weights across pre-treatment periods, the optimization is

$$\mathcal{W}^*_{\text{pre}} := \arg\min_{\mathbf{w}\in\Delta_J} \|\mathbf{y}_1^{\text{pre}} - \mathbf{Y}_0^{\text{pre}}\mathbf{w}\|_2^2, \qquad \widehat{\mathbf{w}} \in \mathcal{W}^*_{\text{pre}}.$$

For any period $t = 1, \ldots, T$, collect the contemporaneous donor outcomes in the column vector

$$\mathbf{y}_{0t} := (Y_{2t}, \ldots, Y_{J+1,t})^{\mathsf{T}} \in \mathbb{R}^J.$$

Good pre-treatment fit means

$$Y_{1t} \approx \widehat{\mathbf{w}}^{\mathsf{T}}\mathbf{y}_{0t}, \qquad t = 1, \ldots, T_0.$$

After treatment, the synthetic outcome estimates the missing counterfactual:

$$\widehat{Y}_{1t}(0) := \widehat{\mathbf{w}}^{\mathsf{T}}\mathbf{y}_{0t} = \sum_{r=1}^{J} \widehat{w}_r Y_{r+1,t}, \qquad t = T_0+1, \ldots, T.$$

The time-specific effect estimate is therefore

$$\widehat{\tau}_{1t} := Y_{1t} - \widehat{Y}_{1t}(0) = Y_{1t} - \widehat{\mathbf{w}}^{\mathsf{T}}\mathbf{y}_{0t}.$$

An average post-treatment effect over periods $T_0+1, \ldots, T$ is

$$\widehat{\overline{\tau}}_1 = \frac{1}{T-T_0} \sum_{t=T_0+1}^{T} \widehat{\tau}_{1t}.$$

This is an effect for the particular treated unit and time period, not automatically a population ATE.

### 3.6.2 Identification under a latent-factor model

One motivation is an interactive factor model for untreated outcomes,

$$Y_{it}(0) = \delta_t + \theta_t^{\mathsf{T}} Z_i + \lambda_t^{\mathsf{T}} \mu_i + \varepsilon_{it},$$

where $Z_i$ contains observed predictors, $\delta_t$ is a common time effect, and $\mu_i$ contains unobserved characteristics whose influence $\lambda_t$ changes over time. A simple DiD model removes only a common time trend. Synthetic control instead uses a long pre-treatment outcome path to match the treated unit's observed predictors and, indirectly, its latent factor loadings.

If $\mathbf{x}_1 \approx \mathbf{X}_0 \widehat{\mathbf{w}}$ and the synthetic control closely tracks the treated unit for many pre-treatment periods, then it may also approximate

$$\delta_t + \theta_t^{\mathsf{T}} Z_1 + \lambda_t^{\mathsf{T}} \mu_1$$

after treatment. This motivates using $\widehat{\mathbf{w}}^{\mathsf{T}} \mathbf{y}_{0t}$ as the untreated counterfactual. It remains an identifying assumption: a close pre-treatment fit does not prove that the relationship will persist after $T_0$.

The donor units must not be affected by the intervention, which is the no-interference requirement. There should be no anticipation before $T_0$, and no other event at $T_0$ should uniquely affect the treated unit. Donors should also be eligible comparison units; a donor with a fundamentally different outcome-generating process may be inappropriate even if an optimization algorithm assigns it weight.

### 3.6.3 Diagnostics and inference

Pre-treatment fit can be summarized by the root mean squared prediction error

$$\mathrm{RMSPE}_{\mathrm{pre}} = \sqrt{\frac{1}{T_0} \sum_{t=1}^{T_0} \left( Y_{1t} - \sum_j w_j^* Y_{jt} \right)^2}.$$

A large pre-treatment RMSPE indicates that the donor pool cannot reproduce the treated unit; post-treatment gaps are then difficult to interpret. The treated and synthetic outcome paths and their gap should always be plotted.

Because there may be only one treated unit, ordinary large-sample standard errors are often inappropriate. A common design-based diagnostic is an in-space placebo study: pretend each donor was treated at $T_0$, construct its own synthetic control, and compare its post-treatment gap with the actual treated unit's gap. One summary statistic is

$$R_i = \frac{\mathrm{RMSPE}_{i,\mathrm{post}}}{\mathrm{RMSPE}_{i,\mathrm{pre}}}.$$

If the treated unit's $R_1$ is unusually large relative to placebo units, its post-treatment divergence is unusual given the donor pool. Placebos with very poor pre-treatment fit should be reported and handled by a stated rule.

Useful robustness checks include leaving out one positively weighted donor at a time, changing the predictor set and pre-treatment window, conducting in-time placebos at dates before the intervention, and reporting the donor weights. These checks reveal whether the conclusion depends on one donor or one modeling choice; they do not validate an implausible causal design.

### 3.6.4 Python implementation

The following example chooses simplex-constrained weights by minimizing pre-treatment squared prediction error. Rows of `y0_pre` and `y0_post` are time periods, and columns are donor units.

```
import numpy as np
from scipy.optimize import minimize

# y1_pre: treated-unit outcomes before treatment, shape (T0,)
# y0_pre: donor outcomes before treatment, shape (T0, J)
# y1_post and y0_post contain the corresponding post-period data.
j = y0_pre.shape[1]

def objective(weights):
    residual = y1_pre - y0_pre @ weights
    return residual @ residual

constraints = {"type": "eq", "fun": lambda w: w.sum() - 1}
bounds = [(0.0, 1.0)] * j
result = minimize(
    objective,
    x0=np.repeat(1.0 / j, j),
    bounds=bounds,
    constraints=constraints,
)
if not result.success:
    raise RuntimeError(result.message)

weights = result.x
synthetic_pre = y0_pre @ weights
synthetic_post = y0_post @ weights
effect_by_period = y1_post - synthetic_post
average_effect = effect_by_period.mean()

pre_rmspe = np.sqrt(np.mean((y1_pre - synthetic_pre) ** 2))
print("Donor weights:", weights)
print("Pre-treatment RMSPE:", pre_rmspe)
print("Post-treatment effects:", effect_by_period)
print("Average post-treatment effect:", average_effect)
```

## 3.7 Uplift Models and Meta-Learners

### 3.7.1 Conditional treatment effects and targeting

Response modeling estimates outcome probabilities or conditional means, whereas uplift modeling estimates differences in expected outcomes under alternative treatments. A high purchase probability under treatment need not imply a large treatment effect. For pre-treatment covariates $X$, binary treatment $A$, and outcome $Y$, define

$$\tau(x) = E[Y(1) - Y(0) \mid X = x].$$

For binary $Y$, this is a difference in probabilities; for revenue or usage it is a difference in conditional means. For example, response probabilities 0.80 versus 0.79 imply uplift 0.01, whereas 0.40 versus 0.20 imply uplift 0.20. The second covariate profile therefore has a larger conditional effect despite

a lower outcome probability under treatment. Uplift modeling commonly emphasizes ranking and targeting by this conditional average treatment effect (CATE). It does not reveal an individual's unobserved pair of potential outcomes.

Under consistency, no interference, conditional exchangeability, and overlap,

$$\{Y(1), Y(0)\} \perp\!\!\!\perp A \mid X, \qquad 0 < e(x) = P(A = 1 \mid X = x) < 1,$$

the CATE is identified as

$$\tau(x) = \mu_1(x) - \mu_0(x), \qquad \mu_a(x) = E[Y \mid X = x, A = a].$$

Randomized experiments provide exchangeability by design; observational applications still need adequate adjustment for confounders. A flexible learner does not remove unmeasured confounding. These methods therefore have different identification requirements from IV or a suitably justified DiD. Meta-learners construct CATE estimators from supervised prediction algorithms. A random forest, for example, can serve as the base learner in multiple constructions. Specialized uplift trees and causal forests provide alternative approaches; the four meta-learners below do not exhaust the class of treatment-effect learning methods.

### 3.7.2 S-learner: a single response model

The S-learner fits a single response model using both $X$ and $A$:

$$\widehat{m} \in \arg\min_{f \in \mathcal{F}} \sum_{i=1}^{n} \{Y_i - f(X_i, A_i)\}^2, \qquad \widehat{\tau}_S(x) = \widehat{m}(x, 1) - \widehat{m}(x, 0).$$

Here squared loss illustrates regression; other appropriate losses can be used. The fitted response surface is evaluated at both treatment values for the same covariate profile. Pooling shares information across arms, but regularization or tree splitting may attenuate treatment dependence and shrink estimated effects toward zero. The designation S refers to the single response model.

### 3.7.3 T-learner: two response models

The T-learner fits separate response models within each treatment arm:

$$\widehat{\mu}_a \in \arg\min_{f \in \mathcal{F}_a} \sum_{i: A_i = a} \{Y_i - f(X_i)\}^2, \qquad a \in \{0, 1\},$$

The estimated conditional effect is

$$\widehat{\tau}_T(x) = \widehat{\mu}_1(x) - \widehat{\mu}_0(x).$$

Separate response models permit arm-specific functional forms, at the cost of reducing the sample available for each fit. Prediction errors from both models contribute to the estimated contrast. Unequal arm sizes and limited overlap can therefore compromise stability.

### 3.7.4 X-learner: imputed treatment-effect regression

The X-learner first estimates $\widehat{\mu}_0$ and $\widehat{\mu}_1$ using arm-specific response models and constructs imputed effect labels:

$$D_i^{(1)} = Y_i - \widehat{\mu}_0(X_i) \quad (A_i = 1), \qquad D_i^{(0)} = \widehat{\mu}_1(X_i) - Y_i \quad (A_i = 0).$$

Each label combines the factual outcome with an estimated counterfactual. For the true conditional mean functions, exchangeability implies

$$E[Y - \mu_0(X) \mid X = x, A = 1] = \mu_1(x) - \mu_0(x) = \tau(x).$$

Separate regressions estimate the conditional means of these labels:

$$\widehat{\tau}_1(x) \approx E[D^{(1)} \mid X = x, A = 1], \qquad \widehat{\tau}_0(x) \approx E[D^{(0)} \mid X = x, A = 0].$$

The final estimate combines the two regressions with a weight $g(x) \in [0, 1]$:

$$\widehat{\tau}_X(x) = g(x)\widehat{\tau}_0(x) + \{1 - g(x)\}\widehat{\tau}_1(x).$$

A common choice is $g(x) = \widehat{e}(x)$. If controls are plentiful and treatment is rare, the control response model may be accurate, so the treated labels $Y - \widehat{\mu}_0(X)$ can be particularly useful. This motivates a larger weight on $\widehat{\tau}_1$ when $e(x)$ is small, although it is not a universal optimal-weight result. X-learning can exploit a simple effect function even when the response functions are complex. Its imputed labels remain noisy estimates, not observed individual treatment effects.

### 3.7.5 DR-learner: doubly robust pseudo-outcome regression

The DR-learner estimates $e, \mu_0, \mu_1$ using sample splitting or cross-fitting to separate nuisance-model training from pseudo-outcome construction. For observation $i$, nuisance predictions from a model excluding its fold are denoted by a superscript $-k(i)$. Construct

$$\begin{aligned}\widehat{\psi}_i = \widehat{\mu}_1^{-k(i)}(X_i) - \widehat{\mu}_0^{-k(i)}(X_i) \\ + \frac{A_i}{\widehat{e}^{-k(i)}(X_i)}\{Y_i - \widehat{\mu}_1^{-k(i)}(X_i)\} \\ - \frac{1 - A_i}{1 - \widehat{e}^{-k(i)}(X_i)}\{Y_i - \widehat{\mu}_0^{-k(i)}(X_i)\}.\end{aligned}$$

An effect model is subsequently fitted to these pseudo-outcomes:

$$\widehat{\tau}_{\mathrm{DR}} \in \arg\min_{f \in \mathcal{F}} \sum_i \{\widehat{\psi}_i - f(X_i)\}^2.$$

The pseudo-outcome augments the fitted conditional mean contrast with inverse-probability-weighted residual corrections. The final regression estimates its conditional expectation as a function of baseline covariates.

To see double robustness, condition on a nuisance training sample and write its fitted functions as $\widetilde{e}, \widetilde{\mu}_0, \widetilde{\mu}_1$. Let $\delta_a(x) = \widetilde{\mu}_a(x) - \mu_a(x)$. Direct conditional expectation yields

$$E[\widetilde{\psi} \mid X = x] - \tau(x) = \{\widetilde{e}(x) - e(x)\}\left\{\frac{\delta_1(x)}{\widetilde{e}(x)} + \frac{\delta_0(x)}{1 - \widetilde{e}(x)}\right\}.$$

Thus the pseudo-outcome has the correct conditional mean if either the propensity function is correct or both response functions are correct. The remaining effect regression must still learn that conditional mean; double robustness is not a guarantee of accurate finite-sample CATE or valid pointwise confidence intervals. Near-zero or near-one propensity scores increase variance. Clipping weights trades stability against bias and does not create overlap where none exists.

DR-learning and DML share orthogonal-score and sample-splitting ideas, but are not interchangeable labels: a common DML analysis estimates a low-dimensional parameter such as an ATE, whereas this DR-learner learns a function $\tau(x)$. Both still require their corresponding identification and estimation conditions.

### 3.7.6 Implementation with CausalML

The following example specifies four meta-learners with a random-forest base regressor using CausalML and scikit-learn. Model specification must be followed by training and independent evaluation.

```
from sklearn.ensemble import RandomForestRegressor
from causalml.inference.meta import (
    BaseSRegressor, BaseTRegressor,
    BaseXRegressor, BaseDRRegressor,
)

def rf():
    return RandomForestRegressor(
        n_estimators=200, min_samples_leaf=30,
        random_state=42, n_jobs=-1,
    )

learners = {
    "S-Learner": BaseSRegressor(learner=rf()),
    "T-Learner": BaseTRegressor(learner=rf()),
    "X-Learner": BaseXRegressor(learner=rf()),
    "DR-Learner": BaseDRRegressor(learner=rf()),
}
```

The dictionary contains four estimators; constructing them does not fit them. The meta-learner can create multiple fitted base models internally. For an RCT, supply the known allocation probabilities where the learner accepts propensity inputs; for observational data, estimate propensities using only appropriate pre-treatment information. Check the installed version's fitting, prediction, and internal splitting interfaces. An outer held-out evaluation set remains necessary even if a learner internally cross-fits nuisance models. For binary outcomes, uplift concerns probabilities rather than thresholded class labels; regression is possible, while classifier-based learners should use probability predictions in the corresponding steps.

### 3.7.7 Evaluation: ranking, calibration, and policy value

Ordinary accuracy or outcome RMSE does not directly measure treatment-effect quality: each user reveals only one potential outcome. In simulations, where the true CATE is known, one can report

$$\mathrm{RMSE}_\tau = \left[\frac{1}{n}\sum_i \{\widehat{\tau}(X_i) - \tau(X_i)\}^2\right]^{1/2}.$$

This quantity cannot generally be computed from real observed outcomes alone. Data partitioning at the randomization-unit level prevents leakage across repeated observations. Model fitting and tuning use development data; the resulting score is fixed before evaluation on an independent test set.

On an independent evaluation RCT, let $s_q(x)$ indicate the top fraction $q$ of users ranked by the frozen score. One population-scaled cumulative gain estimator is

$$\widehat{G}(q) = \frac{1}{n}\sum_i s_q(X_i)\left\{\frac{A_i Y_i}{p_i} - \frac{(1-A_i)Y_i}{1-p_i}\right\},$$

where $p_i$ is the known randomization probability, bounded away from 0 and 1. Conditional on the evaluation features and frozen selection rule, this estimates the corresponding selected effect total divided by $n$. Here $G(1)$ is the full-population average gain; dividing by the selected fraction instead gives the average effect among selected users. Plotting gain against $q$ evaluates targeting. An explicitly defined Qini-style curve is $Q(q) = G(q) - qG(1)$, subtracting random targeting at the same fraction. AUUC integrates an uplift/gain curve, while a Qini area integrates a baseline-adjusted curve; normalization and terminology vary across tools, so report the actual definition. Good ranking does not guarantee calibrated effect magnitudes. Compare predicted effects with randomized effect estimates within prespecified score bins, with uncertainty and sufficient observations. Observational evaluation needs its own justified IPW/DR adjustment, rather than an unadjusted treated–control comparison within each bin.

For a frozen binary targeting policy $\pi(x)$, define

$$V(\pi) = E[\pi(X)Y(1) + \{1 - \pi(X)\}Y(0)].$$

An independent RCT supports the estimator

$$\widehat{V}(\pi) = \frac{1}{n}\sum_i \left\{ \frac{\pi(X_i)A_iY_i}{p_i} + \frac{\{1-\pi(X_i)\}(1-A_i)Y_i}{1-p_i} \right\}.$$

If $Y$ is measured in monetary units and treatment costs $c(x)$, the ideal unconstrained net-benefit rule is $\pi(x) = \mathbf{1}\{\tau(x) > c(x)\}$. With equal per-user costs and a fixed treatment-count budget, rank by uplift; unequal costs require a budget-aware optimization. Policy selection and final value evaluation on the same outcomes lead to optimistic estimates. Use an independent test set and uncertainty estimates respecting the randomization unit; internal random-forest variability is not automatically causal uncertainty.

**References.** For S/T/X meta-learners, see Künzel et al. (2019), *Metalearners for estimating heterogeneous treatment effects using machine learning*, PNAS, doi:10.1073/pnas.1804597116. For implementation interfaces, see the official CausalML documentation, *Meta-Learners Examples* and the API entries for `BaseSRegressor`, `BaseTRegressor`, `BaseXRegressor`, and `BaseDRRegressor`. The PNAS reference concerns S/T/X and should not be read as the source of the DR-learner construction.

## 3.8 Regression Discontinuity (RD)

### 3.8.1 Assignment mechanism and applications

Regression discontinuity exploits a known threshold in a treatment assignment rule. Let $R$ be a pre-treatment *running variable* (also called a score or assignment variable), $c$ the cutoff, $A$ treatment, and $Y$ the outcome. The key comparison is between units just below and just above $c$. For example, a scholarship might be awarded when an entrance score reaches 80. Students scoring 79.9 and 80.1 can be similar in underlying preparation, but their scholarship eligibility differs. RD estimates the jump in the expected later outcome at 80, after allowing the outcome to vary smoothly with the score on either side. Comparing all scholarship recipients with all nonrecipients would generally confound the scholarship with initial ability.

Useful settings include scholarship score cutoffs, income-based program eligibility, age-based policy eligibility, and platform benefits triggered by a pre-existing score. The threshold must actually

change treatment or its probability. A threshold selected after inspecting outcomes does not establish an RD design. An age cutoff may activate several policies simultaneously; in that case the outcome jump cannot automatically be attributed to just one. These examples are candidate designs, not guarantees of valid identification.

### 3.8.2 Sharp RD: identification by continuity

In a sharp design, the rule determines treatment exactly:

$$A_i = \mathbf{1}\{R_i \geq c\}, \qquad Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0).$$

Define $m_a(r) = E[Y_i(a) \mid R_i = r]$. Suppose both $m_0$ and $m_1$ are continuous at $c$, with observations approaching the cutoff from both sides, and assume consistency and an appropriate no-interference condition. For $r < c$, the observed mean equals $m_0(r)$; for $r \geq c$, it equals $m_1(r)$. Consequently,

$$\begin{aligned}
\lim_{r \to c^+} E[Y \mid R = r] &= \lim_{r \to c^+} m_1(r) = m_1(c), \\
\lim_{r \to c^-} E[Y \mid R = r] &= \lim_{r \to c^-} m_0(r) = m_0(c), \\
\tau_{\text{SRD}} &\equiv \lim_{r \to c^+} E[Y \mid R = r] - \lim_{r \to c^-} E[Y \mid R = r] \\
&= m_1(c) - m_0(c) = E[Y(1) - Y(0) \mid R = c].
\end{aligned}$$

For a continuous score, the conditional means at the cutoff are understood through their continuous limits, not as a sample mean of observations exactly equal to $c$. Identification follows from continuity of both potential outcome regression functions combined with discontinuity of the assignment rule at the threshold. It does not require the entire population on one side to resemble the entire population on the other side.

**Target estimand** Sharp RD identifies a *local average effect at the cutoff*, a CATE conditional on $R = c$. It is generally neither the population ATE nor the ATT among all treated units. Extrapolation to scores far from $c$ needs additional assumptions. Unlike adjustment methods based on overlap, sharp RD has $P(A = 1 \mid R = r) \in \{0, 1\}$: ordinary positivity conditional on $R$ fails. Continuity across the boundary, rather than observing both arms at the same score, supplies identification. The continuity-based framework does not assert exact randomization within a finite window; such an assertion requires a separate local-randomization assumption.

### 3.8.3 Local linear estimation and bandwidth

Set $r_i = R_i - c$ and retain observations near the cutoff. A local linear fit with separate slopes solves

$$(\widehat{\alpha}, \widehat{\tau}, \widehat{\beta}, \widehat{\gamma}) = \arg \min_{\alpha, \tau, \beta, \gamma} \sum_{i : |r_i| < h} K(r_i/h) \{Y_i - \alpha - \tau A_i - \beta r_i - \gamma A_i r_i\}^2.$$

For the triangular kernel, $K(u) = (1 - |u|)\mathbf{1}\{|u| < 1\}$. The left fitted line has intercept $\alpha$; the right fitted line has intercept $\alpha + \tau$. Thus their estimated jump at the cutoff is $\widehat{\tau}$. Centering at $c$ makes this coefficient directly interpretable. The interaction allows different slopes; imposing the same slope can create unnecessary specification error.

The bandwidth $h$ controls the bias–variance tradeoff: a narrow neighborhood is more local but contains fewer observations, whereas a wide neighborhood requires a more demanding approximation to the response curves. Prefer local low-order polynomials with principled bandwidth selection

to a global high-degree polynomial. At an MSE-oriented bandwidth, ordinary regression standard errors can miss smoothing bias. Use RD-specific robust bias-corrected inference or another justified RD inference procedure; adding heteroskedasticity- robust errors to an arbitrary global regression does not solve this problem.

### 3.8.4 Fuzzy RD: a local instrumental-variable estimand

Sometimes the cutoff changes eligibility but not everyone complies. Let $Z_i = \mathbf{1}\{R_i \geq c\}$ denote eligibility, and let $A_i(z)$ denote treatment receipt under eligibility $z$. If treatment probability jumps,

$$\Delta_A = \lim_{r \to c^+} E[A \mid R = r] - \lim_{r \to c^-} E[A \mid R = r] \neq 0,$$

define the outcome jump $\Delta_Y$ analogously. Under the relevant continuity conditions for potential treatment receipt and outcomes, exclusion (eligibility affects the outcome only through receipt), and monotonicity $A_i(1) \geq A_i(0)$, the local IV argument gives

$$\begin{aligned} \Delta_Y &= E[(A_i(1) - A_i(0))(Y_i(1) - Y_i(0)) \mid R_i = c], \\ \Delta_A &= E[A_i(1) - A_i(0) \mid R_i = c] \\ &= P(A_i(1) > A_i(0) \mid R_i = c), \\ \tau_{\text{FRD}} = \frac{\Delta_Y}{\Delta_A} &= E[Y_i(1) - Y_i(0) \mid A_i(1) > A_i(0), R_i = c]. \end{aligned}$$

This is the effect for *compliers near the cutoff*. For example, an outcome jump of 4 and a treatment-probability jump of 0.5 imply a fuzzy RD estimate of 8 outcome units, subject to these assumptions. A small first-stage jump makes the ratio unstable. Eligibility itself may matter independently of treatment receipt, in which case the exclusion restriction is questionable.

### 3.8.5 Diagnostics and limits

- Plot outcome and treatment receipt against the running variable, using separate fits on either side. A binned picture is exploratory evidence, not a substitute for estimation and inference.
- Investigate precise score manipulation and sorting around the cutoff. A density test and institutional knowledge can reveal concerns; passing a density test does not prove that potential outcomes are continuous.
- Examine pre-treatment covariates and placebo outcomes for discontinuities. Check whether any other policy changes at the same cutoff. Covariate adjustment cannot automatically rescue a fundamentally invalid threshold comparison.
- Assess bandwidth sensitivity and data support on both sides. Rounded or discrete scores, few distinct score values, and gaps near the cutoff can undermine the continuous-score approximation and require different inference.
- Avoid post-treatment running variables, outcome-driven cutoff selection, and interpreting a local effect as a population-wide effect. If spillovers cross the threshold, the standard no-interference interpretation may fail.

### 3.8.6 Python and R implementations

The following sharp-RD simulation has a true cutoff effect of 2 and different slopes on the two sides. The `rdrobust` package provides local-polynomial estimation, bandwidth selection, and robust bias-corrected inference. Inspect the robust inference output, not only the conventional interval. The two languages use different random-number generators and need not give identical finite-sample results.

```
# Python: pip install numpy rdrobust
import numpy as np
from rdrobust import rdrobust

rng = np.random.default_rng(20260923)
r = rng.uniform(-1, 1, 4000)  # running variable; cutoff = 0
a = (r >= 0).astype(int)
y = 1 + 0.5*r + 2*a + 0.3*a*r + rng.normal(size=r.size)
fit = rdrobust(y=y, x=r, c=0, p=1, kernel="triangular")
print(fit)

# For fuzzy RD, a_received must be ACTUAL treatment receipt:
# fit = rdrobust(y=y, x=r, c=0, fuzzy=a_received, p=1)
```

```
# R: install.packages("rdrobust")
library(rdrobust)
set.seed(20260923)
r <- runif(4000, -1, 1)
a <- as.integer(r >= 0)
y <- 1 + 0.5*r + 2*a + 0.3*a*r + rnorm(length(r))
fit <- rdrobust(y = y, x = r, c = 0, p = 1,
                kernel = "triangular")
summary(fit)

# Fuzzy RD: supply actual treatment receipt, not eligibility.
# fit <- rdrobust(y = y, x = r, c = 0,
#                 fuzzy = a_received, p = 1)
```

For identification and practice, see Imbens and Lemieux (2008), *Regression discontinuity designs: A guide to practice*, Journal of Econometrics, 142(2), 615–635; and Lee and Lemieux (2010), *Regression Discontinuity Designs in Economics*, Journal of Economic Literature, 48(2), 281–355. Software documentation is maintained at `rdpackages.github.io` under `rdrobust`.

# 4 Design and Analysis of Randomized Experiments

## 4.1 CUPED

CUPED (Controlled-experiment Using Pre-Experiment Data) is a covariate-adjustment method for improving precision in randomized experiments. It removes an outcome component predictable from pre-treatment measurements, such as pre-experiment expenditure when the outcome is expenditure during follow-up. Identification is supplied by randomization; the adjustment reduces sampling variance and does not resolve unmeasured confounding in observational studies.

### 4.1.1 Adjusted outcome and treatment-effect estimator

Let $A_i \in \{0,1\}$ be randomized treatment, $Y_i$ the experiment-period outcome, and $X_i$ a scalar pre-experiment measurement. Assume independent experimental units, consistency, no interference, and

$$A_i \perp\!\!\!\perp \{Y_i(0), Y_i(1), X_i\}, \qquad 0 < \Pr(A_i = 1) < 1.$$

For a fixed coefficient $\theta$ and a common centering constant $c$, define

$$Y_i^{\mathrm{adj}} = Y_i - \theta(X_i - c).$$

One can take $c = \mathbb{E}[X]$ or the pooled sample mean $\overline{X}$. Writing $\overline{Y}_a$ and $\overline{X}_a$ for the means in arm $a$, the adjusted difference is

$$\begin{aligned}\widehat{\tau}_{\mathrm{CUPED}}(\theta) &= \overline{Y}_1^{\mathrm{adj}} - \overline{Y}_0^{\mathrm{adj}} \\ &= (\overline{Y}_1 - \overline{Y}_0) - \theta(\overline{X}_1 - \overline{X}_0).\end{aligned}$$

The common centering constant cancels exactly. Randomization implies that the baseline difference has expectation zero, so, for fixed $\theta$,

$$\mathbb{E}[\widehat{\tau}_{\mathrm{CUPED}}(\theta)] = \mathbb{E}[Y(1) - Y(0)] = \tau_{\mathrm{ATE}}.$$

Thus CUPED corrects chance baseline imbalance without changing the target effect. This exact unbiasedness argument treats $\theta$ as fixed (or learned independently); estimating it on the same experiment generally gives asymptotic validity under regularity conditions, not automatic exact finite-sample unbiasedness.

### 4.1.2 Optimal coefficient and variance reduction

For a square-integrable pair $(X, Y)$ with $\mathrm{Var}(X) > 0$,

$$\begin{aligned}\mathrm{Var}\{Y - \theta(X - c)\} &= \mathrm{Var}(Y) + \theta^2 \mathrm{Var}(X) - 2\theta \,\mathrm{Cov}(X, Y), \\ \frac{\partial}{\partial\theta} \mathrm{Var}\{Y - \theta(X - c)\} &= 2\theta \,\mathrm{Var}(X) - 2\,\mathrm{Cov}(X, Y).\end{aligned}$$

Setting the derivative to zero yields

$$\theta^* = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad \mathrm{Var}(Y^{\mathrm{adj}}) = \mathrm{Var}(Y)(1 - \rho^2),$$

where $\rho = \mathrm{Corr}(X, Y)$. The stronger the baseline prediction, the larger the reduction. For $\rho = 0.8$, the adjusted variance is 0.36 times the original variance, while the standard deviation is 0.6 times the original. A 64% reduction in variance is not a 64% reduction in standard error.

The optimal coefficient for a two-arm difference should be distinguished from that for a single outcome. With fixed arm sizes $n_1, n_0$, define $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_a^2 = \mathrm{Var}\{Y(a)\}$, and $c_a = \mathrm{Cov}\{X, Y(a)\}$. Under independent sampling,

$$\mathrm{Var}\{\widehat{\tau}(\theta)\} = \frac{\sigma_1^2 + \theta^2\sigma_X^2 - 2\theta c_1}{n_1} + \frac{\sigma_0^2 + \theta^2\sigma_X^2 - 2\theta c_0}{n_0}.$$

Consequently,

$$\theta^*_{\mathrm{diff}} = \frac{c_1/n_1 + c_0/n_0}{\sigma_X^2(1/n_1 + 1/n_0)}.$$

When both arms have the same baseline–outcome covariance, this reduces to the familiar covariance-to-variance ratio. Under a constant additive effect $Y(1) = Y(0) + \tau$, both arms have the same variance and covariance, and the variance of the treatment-effect estimator also decreases by the factor $1 - \rho^2$, with $\rho = \mathrm{Corr}\{X, Y(0)\}$. This factor is not a universal guarantee for heterogeneous effects or estimated coefficients.

For several baseline covariates, let $\mathbf{X}_i \in \mathbb{R}^p$ be a column vector, with nonsingular covariance matrix $\mathbf{\Sigma}_X \in \mathbb{R}^{p \times p}$. The analogous adjustment is

$$Y_i^{\mathrm{adj}} = Y_i - \boldsymbol{\theta}^{\mathsf{T}}(\mathbf{X}_i - \mathbf{c}), \qquad \boldsymbol{\theta}^* = \mathbf{\Sigma}_X^{-1} \operatorname{Cov}(\mathbf{X}, Y).$$

The resulting single-outcome variance is

$$\operatorname{Var}(Y) - \operatorname{Cov}(\mathbf{X}, Y)^{\mathsf{T}} \mathbf{\Sigma}_X^{-1} \operatorname{Cov}(\mathbf{X}, Y).$$

### 4.1.3 Relation to regression adjustment and DiD

A practical linear implementation is ANCOVA:

$$Y_i = \alpha + \tau A_i + \theta(X_i - \overline{X}) + \varepsilon_i.$$

The OLS normal equations imply exactly

$$\widehat{\tau}_{\mathrm{OLS}} = (\overline{Y}_1 - \overline{Y}_0) - \widehat{\theta}_{\mathrm{OLS}}(\overline{X}_1 - \overline{X}_0).$$

Thus this regression produces a CUPED-form estimate using a jointly estimated common slope. Its slope need not equal the sample covariance-to-variance ratio from a separate pooled regression that omits treatment.

If $X$ is the pre-period value of the same outcome, setting $\theta = 1$ gives the numerical two-period DiD contrast. However, their causal justifications differ: CUPED here uses randomization and learns a noise-reduction coefficient, whereas observational DiD uses a parallel-trends assumption. Adding a baseline metric to an observational comparison does not automatically identify an ATE.

### 4.1.4 Practical requirements and Python example

Eligible adjustment covariates are recorded before exposure and are unaffected by treatment. Adjustment for post-treatment engagement or other mediators can change the estimand or introduce selection bias. The metric, baseline window, missing-data procedure, and analysis population are prespecified. Exclusion of users without baseline history changes the analysis population and requires explicit justification. Flexible adjustment models may require sample splitting or cross-fitting to limit overfitting.

The example below uses one row per independently randomized user, a continuous outcome, and a balanced experiment. It implements the common-slope regression version and reports heteroskedasticity-robust confidence intervals. If treatment is randomized by cluster, uncertainty should instead respect that clustering. With heterogeneous slopes or unequal allocation, consider centered treatment–covariate interactions and a corresponding marginal ATE estimator.

```
import numpy as np
import pandas as pd
import statsmodels.formula.api as smf
```

```
rng = np.random.default_rng(42)
n = 4000
x = rng.normal(size=n)  # pre-experiment user metric
a = rng.binomial(1, 0.5, size=n)  # randomized assignment
y = 10 + 2 * a + 3 * x + rng.normal(size=n)
df = pd.DataFrame({"Y": y, "A": a, "X": x})
df["Xc"] = df["X"] - df["X"].mean()

raw = smf.ols("Y ~ A", data=df).fit(cov_type="HC3")
cuped = smf.ols("Y ~ A + Xc", data=df).fit(cov_type="HC3")
theta = cuped.params["Xc"]
df["Y_adj"] = df["Y"] - theta * df["Xc"]
means = df.groupby("A")["Y_adj"].mean()
assert np.isclose(means.loc[1] - means.loc[0],
                  cuped.params["A"])

for name, fit in [("Raw", raw), ("CUPED/ANCOVA", cuped)]:
    print(name, "ATE:", fit.params["A"])
    print("SE:", fit.bse["A"])
    print("95% CI:", fit.conf_int().loc["A"].to_numpy())
print("Estimated theta:", theta)
print("SE ratio:", cuped.bse["A"] / raw.bse["A"])
```

The simulated treatment effect is 2. The adjusted and unadjusted estimators target this same effect, but the baseline metric explains much of the outcome variation, making the adjusted estimate more precise. For background, see Deng, Xu, Kohavi, and Walker (2013), *Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data*, WSDM, doi:10.1145/2433396.2433413.

## 4.2 Sample Ratio Mismatch (SRM)

### 4.2.1 Definition and motivation

Sample ratio mismatch is a discrepancy between observed experimental group sizes and the configured allocation proportions that is too large to be readily explained by the randomization mechanism. It is a diagnostic of assignment or data quality, not a test of treatment effectiveness. A planned 50:50 experiment need not contain exactly equal numbers: ordinary random variation is expected under independent assignment.

For example, suppose 5,000 users are assigned to each arm, but 1,000 treatment users disappear from the analysis because the new page fails to log their visits. The remaining sample has a 5,000:4,000 ratio. If failures are associated with non-purchase, conversion among the remaining treatment users can be artificially inflated. SRM flags the discrepancy, but does not identify its cause or the direction of outcome bias. Conversely, passing an SRM check does not prove that an experiment is free of other data problems.

Count distinct randomization units, using treatment-independent eligibility rules. In a user-randomized experiment, order counts or click counts need not follow the allocation ratio because treatment can change those behaviors. Likewise, unequal student counts in a school-randomized experiment are not automatically evidence of faulty school assignment.

### 4.2.2 Testing for SRM

Suppose $K \geq 2$ arms have prespecified probabilities $p_k > 0$, with $\sum_{k=1}^{K} p_k = 1$. Let $O_k$ be the number of observed distinct units in arm $k$, and $n = \sum_k O_k$. Under independent assignment with constant probabilities and treatment-independent inclusion, the null model is

$$H_0 : \quad (O_1, \ldots, O_K) \sim \text{Multinomial}(n; p_1, \ldots, p_K).$$

Define the expected counts $E_k = np_k$. Pearson's statistic is

$$Q = \sum_{k=1}^{K} \frac{(O_k - E_k)^2}{E_k}, \qquad Q \overset{H_0}{\approx} \chi^2_{K-1}.$$

The approximation requires sufficiently large expected counts. The tail probability is $\Pr(\chi^2_{K-1} \geq Q_{\text{obs}})$. A small value flags incompatibility with the null allocation model; it is not the probability that the implementation contains a bug. For two arms and small samples, an exact binomial test can instead use $O_1 \sim \text{Binomial}(n, p_1)$.

For a 50:50 allocation with observed counts $(6000, 4000)$,

$$E_1 = E_2 = 5000, \qquad Q = \frac{1000^2}{5000} + \frac{(-1000)^2}{5000} = 400,$$

giving overwhelming evidence against the null. In contrast, counts $(5020, 4980)$ give $Q = 0.16$ and a $p$-value of approximately 0.689.

Prespecify the alert threshold and monitoring schedule. For example, $\alpha_{\text{SRM}} = 0.001$ can be an operational choice, but is not a universal rule. Repeated daily tests or searches across many segments require appropriate monitoring or multiplicity handling; their unadjusted $p$-values are not a single confirmatory test. Report the counts, planned proportions, absolute proportion differences, and $p$-value together.

The reference distribution must respect the actual design. Fixed quotas, blocked randomization, or cluster assignment need design-specific checks. If allocation probabilities change over time, inspect counts within the corresponding allocation periods; do not compare all users with an obsolete 50:50 plan. Summing period-specific expected counts alone does not generally justify the ordinary constant-probability multinomial reference distribution.

```
import numpy as np
from scipy.stats import chisquare, binomtest

# Distinct randomized users: control, then treatment.
observed = np.array([6000, 4000])
planned = np.array([0.5, 0.5])
expected = observed.sum() * planned
test = chisquare(f_obs=observed, f_exp=expected)
alpha_srm = 0.001  # illustrative, prespecified alert threshold
print("Observed:", observed, "Expected:", expected)
print("Chi-square:", test.statistic, "p-value:", test.pvalue)
print("SRM alert:", test.pvalue < alpha_srm)

# Exact alternative for two arms with independent assignment.
exact = binomtest(int(observed[1]), n=int(observed.sum()),
                  p=float(planned[1]), alternative="two-sided")
print("Exact binomial p-value:", exact.pvalue)
```

### 4.2.3 Common causes and corresponding remedies

**Assignment errors.** Incorrect hash buckets, unstable user identifiers, or conflicting configuration can alter assignment. Audit the configuration history and assignment logs, verify persistent identifiers, and repair the allocation mechanism. An A/A test can help check the repaired pipeline.

**Execution or exposure differences.** Redirects, crashes, or treatment-dependent loading can cause users to disappear from exposure logs. Compare assignment and exposure counts, retain reliable assignment records, and repair instrumentation. A real treatment effect on exposure should be reported as part of the experiment, rather than removed by restricting analysis to successfully exposed users.

**Logging and data processing.** Variant-specific missing events, delayed logs, duplicates, and incorrect joins can change group sizes. Compare source records with each pipeline stage, use consistent time windows, validate join keys and cardinalities, and deduplicate at the randomization-unit level. Reprocess data only when reliable source records permit reconstruction.

**Post-treatment filtering.** Restricting analysis to clickers, purchasers, or returning users can create unequal inclusion because treatment affects the filter. Prefer the prespecified eligible, assigned population for an intention-to-treat analysis. A triggered analysis requires a justified trigger definition that preserves causal comparability; conditioning on an affected behavior is not an automatic solution.

**Incorrect expected ratio or time window.** A planned traffic ramp, staggered enrollment, or inconsistent cutoff times can make the assumed ratio wrong. Reconstruct expected allocation from configuration logs and analyze appropriate periods or strata. Do not redefine the expected ratio to equal the observed ratio merely to pass.

### 4.2.4 Practical investigation and resolution

Locate the first stage where proportions depart from the applicable plan: eligibility, assignment, exposure, logging, or the final analysis table. Inspect treatment-independent segments such as preexisting device type and enrollment date to localize failures. Record whether the problem is only a misstated reference ratio or actual differential loss of observations.

After fixing a recoverable pipeline error, rebuild the data from trustworthy records and repeat the quality checks. If observations or outcomes cannot be recovered, document the missingness and its possible causal implications; rerunning the experiment after repair may be necessary. Restoring an assignment denominator does not by itself recover missing outcomes.

Discarding users solely to equalize arm sizes does not resolve the underlying allocation or selection problem. Neither IPW nor CUPED automatically corrects unknown selection or logging bias. Causal interpretation of affected results requires an explanation of the discrepancy and justification of the resulting analysis population and observation process.

For the diagnostic taxonomy, see Fabijan et al. (2019), *Diagnosing Sample Ratio Mismatch in Online Controlled Experiments: A Taxonomy and Rules of Thumb for Practitioners*, KDD, doi:10.1145/3292500.3330722.

## 4.3 Randomization: Deterministic Hashing and Units of Analysis

### 4.3.1 Statistical goal: a reproducible random assignment

Let $i = 1, \ldots, n$ index eligible users, $e$ identify an experiment, and $A_{ie} \in \{0, 1\}$ denote assignment. For treatment probability $p$, the ideal Bernoulli design draws independent $U_{ie} \sim \text{Uniform}[0, 1)$, independently of pre-treatment characteristics and potential outcomes, and sets

$$A_{ie} = \mathbf{1}\{U_{ie} < p\}, \qquad \Pr(A_{ie} = 1 \mid X_i, Y_i(0), Y_i(1)) = p.$$

Repeated observations of a user retain the same treatment assignment. Deterministic hashing provides a reproducible pseudorandom implementation: a fixed identifier and experiment configuration yield an invariant assignment across visits.

**Randomness and reproducibility** A fixed hash of fixed IDs is a deterministic function, not a literal random variable. One can select a random experiment key $K_e$ before assignment, independently of user data, and then hold it fixed. Under an ideal random-function model, distinct IDs produce independent uniform outputs over the random key; a cryptographic keyed hash is a practical pseudorandom approximation to that model, not a proof of exact statistical independence. Conditional on the realized key and ID, assignment has no remaining randomness. Randomization-based reasoning concerns the assignment mechanism before fixing its random seed, just as with a seeded pseudorandom number generator. Selection of a seed based on observed treatment effects invalidates the specified randomization mechanism. Hash-based assignment does not address treatment-dependent eligibility or missing outcomes.

### 4.3.2 From a stable ID to an integer, a bucket, and a treatment

Choose a canonical, unambiguous encoding enc and define

$$\begin{aligned} m_{ie} &= \text{enc}(\text{version}, e, \text{namespace}, \text{user ID}_i), \\ d_{ie} &= \text{HMAC-SHA256}(K_e, m_{ie}), \\ h_{ie} &= \text{Int}_{\text{big endian}}(d_{ie}[0{:}8]), \qquad M = 2^{64}, \\ u_{ie} &= h_{ie}/M \in \{0, 1/M, \ldots, (M-1)/M\}. \end{aligned}$$

The digest is a byte string; taking its first eight bytes and interpreting them as an unsigned big-endian integer gives $0 \le h_{ie} < M$. The hash maps an identifier to a reproducible numerical value without using the individual's predicted treatment response. HMAC is one explicit choice, not a requirement that all experimentation platforms use a cryptographic algorithm.

To avoid floating-point boundary differences, write $p = a/b$ and assign with integer arithmetic:

$$T = \left\lfloor \frac{aM}{b} \right\rfloor, \qquad A_{ie} = \mathbf{1}\{h_{ie} < T\}.$$

Under uniform hash output, $\Pr(A_{ie} = 1) = T/M$, which differs from $p$ by less than $1/M$. For $p = 1/2$, $T = 2^{63}$ and the probability is exactly $1/2$ under that model. This discretization error is negligible, but its definition is useful for consistent implementations across languages.

**Bucket-based allocation** The common shorthand is

$$B_{ie} = h_{ie} \bmod 100 \in \{0, \ldots, 99\}, \qquad A_{ie} = \mathbf{1}\{B_{ie} < 50\}.$$

For example, an illustrative hash integer $h = 12345678$ gives bucket 78 and control; $h = 12345623$ gives bucket 23 and treatment. These small integers illustrate modulo arithmetic, not actual HMAC results. More generally, if $M = qB + r$ with $0 \le r < B$, modulo $B$ gives

$$\Pr(h \bmod B = j) = \begin{cases} (q+1)/M, & 0 \le j < r, \\ q/M, & r \le j < B. \end{cases}$$

Thus modulo is exactly uniform only if $B$ divides $M$. With 64 bits and 100 buckets the bias is tiny. Exact uniform buckets can instead reject $h \ge B\lfloor M/B \rfloor$ and rehash using an incrementing counter, under the ideal independent-output model. For ordinary arm assignment the direct integer threshold above is simpler and provides much finer resolution than 100 buckets. This discretization effect is negligible relative to the allocation discrepancies ordinarily investigated as SRM.

### 4.3.3 Bernoulli allocation and realized sample sizes

Under ideal independent Bernoulli assignment,

$$N_1 = \sum_{i=1}^{n} A_{ie} \sim \text{Binomial}(n, p), \quad E[N_1] = np, \quad \text{Var}(N_1) = np(1-p).$$

At $n = 10{,}000$ and $p = 1/2$, the standard deviation is 50 users, so a rough 95% range is $5000 \pm 98$. Hashing allocates probabilities, not fixed quotas. These binomial statements assume the ideal assignment model; a fixed realized hash configuration does not itself generate repeated random draws.

For an exactly fixed treatment size $n_1$, a different design is needed: randomly permute a fixed eligible roster and treat its first $n_1$ users. Ranking pseudorandom hash scores approximates that design (handle ties explicitly), but cannot keep assignments stable if the roster is continually changed and ranks are recomputed. Fixed-size assignments are dependent and need their own randomization reference distribution. For blocked randomization, form strata using pre-treatment variables and randomize within each stratum; equal probabilities within strata alone do not force exact balance within each stratum.

### 4.3.4 Python implementation and assignment protocol

Do not use Python's built-in `hash()` for a persistent, cross-service assignment protocol. Define the byte encoding, hash algorithm, integer width, byte order, namespace, key, and allocation rule explicitly. The following example uses only the Python standard library. All identifiers are strings; no implicit trimming, lowercasing, or conversion between `"001"` and `"1"` is performed.

```
import hashlib
import hmac

def encode_fields(*fields):
    # Length-prefixed UTF-8 avoids ambiguous concatenation.
    result = bytearray()
    for field in fields:
        if not isinstance(field, str):
            raise TypeError("IDs must be canonical strings")
        raw = field.encode("utf-8")
        result.extend(len(raw).to_bytes(4, "big"))
```

```
        result.extend(raw)
    return bytes(result)

def score(user_id, experiment_id, key, namespace="arm"):
    msg = encode_fields("v1", experiment_id, namespace,
                        user_id)
    digest = hmac.new(key, msg, hashlib.sha256).digest()
    return int.from_bytes(digest[:8], "big")

def assign(user_id, experiment_id, key, a=1, b=2):
    if not (isinstance(a, int) and isinstance(b, int)
            and b > 0 and 0 <= a <= b):
        raise ValueError("Require integers 0 <= a <= b")
    threshold = (a * (1 << 64)) // b
    return int(score(user_id, experiment_id, key)
               < threshold)

# Demo only. In a real experiment, generate once with
# secrets.token_bytes(32), persist securely, and reuse.
key = b"tutorial-key-fixed-for-reproducibility"
u = "user-001"
a1 = assign(u, "feed-exp-2026", key)
assert a1 == assign(u, "feed-exp-2026", key)
assert assign(u, "feed-exp-2026", key, 0, 1) == 0
assert assign(u, "feed-exp-2026", key, 1, 1) == 1
assert encode_fields("ab", "c") != encode_fields("a", "bc")
print("Assignment (1=treatment):", a1)
print("Treatment count:", sum(
    assign(str(i), "feed-exp-2026", key)
    for i in range(10000)))
```

An experiment ID separates experiments; a namespace separates purposes such as enrollment and arm assignment. These give pseudorandom separation, not a mathematical proof of independence for a particular fixed hash function. Cross-language implementations should share fixed input/output test vectors. The same ID on two devices gets the same assignment only if both devices actually resolve to that same stable ID.

**Traffic ramps and persistent assignment.** Using the same score and changing $p$ from 0.5 to 0.6 switches users with $u \in [0.5, 0.6)$ from control to treatment. Determinism does not prevent this. A common design separates enrollment $R_i$ from arm $A_i$:

$$R_i = \mathbf{1}\{u_i^{\text{enroll}} < q\}, \qquad A_i = \mathbf{1}\{u_i^{\text{arm}} < p\},$$

with separate hash namespaces and fixed $p$. Increasing enrollment fraction $q$ then admits more users while preserving existing arm assignments. Persist initial assignment and its configuration version when the protocol requires stickiness, and specify how late enrollment and exposure windows enter analysis. Changing a key, ID definition, or encoding also changes assignments and should not happen silently during an experiment.

### 4.3.5 Randomization unit versus analysis unit

Suppose users are randomized but each user generates multiple session rows. Let $A_i$ be the user's assignment and $Y_{ij}$ the outcome of session $j$. All observations for a user share $A_i$ and therefore do

not represent independent treatment assignments. A simple model with fixed $m$ sessions per user illustrates the variance issue:

$$Y_{ij} = \alpha + \tau A_i + b_i + \varepsilon_{ij}, \qquad \rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_\varepsilon^2}.$$

If $n$ independent users each contribute $m$ observations, then

$$\mathrm{Var}(\bar{Y}) = \frac{\sigma_b^2}{n} + \frac{\sigma_\varepsilon^2}{nm} = \frac{\sigma_b^2 + \sigma_\varepsilon^2}{nm}\{1 + (m-1)\rho\}.$$

Relative to an independent-row calculation, the design effect is $1 + (m-1)\rho$ and the illustrative effective sample size is $nm/\{1 + (m-1)\rho\}$. With $m = 10$ and $\rho = 0.2$, the variance multiplier is 2.8 and the standard-error multiplier is $\sqrt{2.8} \approx 1.67$. This equal-size, exchangeable-correlation illustration is not a universal formula for unequal clusters or treatment-dependent numbers of sessions.

**Estimands and variance estimation.** For total revenue per randomized user over a fixed follow-up period, define $S_i(a)$ as that user's total under assignment $a$, including zero for genuine zero activity (not for missing telemetry). The target and estimator are

$$\tau_{\text{user}} = E[S_i(1) - S_i(0)], \qquad \widehat{\tau}_{\text{user}} = \frac{\sum_i A_i S_i}{N_1} - \frac{\sum_i (1 - A_i) S_i}{N_0}.$$

Aggregate to one row per user and use user-level inference. With session rows, cluster-robust uncertainty at the user level or a user-level bootstrap can account for within-user dependence, provided enough independent users and appropriate regularity conditions. Cluster-robust errors do not automatically change a session-weighted coefficient into a user-weighted estimand.

In particular, the ratio

$$\widehat{r}_a = \frac{\sum_{i:A_i=a} S_i}{\sum_{i:A_i=a} M_i}$$

weights users by their number of sessions $M_i$ and differs from the average of user-specific ratios. If treatment changes session counts, this is a ratio of policy-specific totals, not automatically an effect on a fixed population of sessions. Preserve each user's numerator and denominator together in a bootstrap or delta-method calculation, and do not condition on post-treatment activity without justifying the resulting estimand.

For household-, school-, or geographic-cluster assignment, inference must respect those randomized clusters, not merely individual rows. A large number of users within a small number of regions does not create a large number of independent treatment assignments; small-cluster methods or randomization inference matching the actual design may be needed. Finally, cross-user spillovers are an identification issue as well as a variance issue: clustering standard errors does not restore no interference. Choose a design that addresses plausible spillovers and define the relevant causal effect before assignment.

**Implementation and reporting considerations.** A reproducible allocation protocol specifies the stable randomization identifier, protocol version, allocation probability, and source of pseudorandomness. Implementation preserves assignments across visits and traffic ramps and records assignment before exposure. SRM assessment and variance estimation are conducted at levels consistent with the assignment design and target estimand. For implementation background, see Optimizely's official documentation, *How bucketing works in Optimizely Feature Experimentation*, and the Python standard-library documentation for `hmac` and `hashlib`.

## 4.4 Type I/II Errors, Statistical Power, and Test Statistics

### 4.4.1 Definitions of Type I and Type II errors

A hypothesis test specifies a null hypothesis $H_0$, an alternative $H_1$, and a decision rule. For an A/B test, a common choice is

$$H_0 : \tau = 0, \qquad H_1 : \tau \neq 0,$$

where $\tau$ is the causal effect on a prespecified metric. For an interference test, $H_0$ instead states that there is no cross-unit interference; it need not state that the direct treatment effect is zero. The two possible errors are

| | $H_0$ is true | $H_1$ is true |
|---|---|---|
| Reject $H_0$ | Type I error | Correct detection |
| Do not reject $H_0$ | Correct non-rejection | Type II error |

A Type I error is rejection of a true null hypothesis. A Type II error is non-rejection under a specified alternative. Failure to reject is not a proof of no effect. Demonstrating that an effect is negligibly small requires a prespecified equivalence margin and an appropriate equivalence test or confidence interval.

Let $\varphi(D) \in \{0, 1\}$ indicate rejection. A test of level $\alpha$ satisfies

$$\sup_{\theta \in \Theta_0} \Pr_\theta\{\varphi(D) = 1\} \leq \alpha.$$

The supremum matters for a composite null with unknown nuisance parameters. For a particular alternative $\theta \in \Theta_1$, define

$$\beta(\theta) = \Pr_\theta\{\varphi(D) = 0\}, \qquad \text{Power}(\theta) = 1 - \beta(\theta).$$

Thus $\alpha$ controls false alarms, whereas power measures the probability of detecting a specified true signal. Power is a function of the effect, sample size, noise, design, and test; it is not one universal number. It is also not the posterior probability that $H_1$ is true.

### 4.4.2 Deriving power for a difference in means

Suppose there are $n_1$ independent treated units and $n_0$ independent control units. Let the arm means be $\mu_1, \mu_0$, arm variances be $\sigma_1^2, \sigma_0^2$, and $\tau = \mu_1 - \mu_0$. Then

$$\widehat{\tau} = \bar{Y}_1 - \bar{Y}_0, \qquad s^2 = \text{Var}(\widehat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

With normal outcomes and known variances, the following normal distribution is exact; with estimated variances and suitable large samples it is an approximation:

$$Z = \frac{\widehat{\tau}}{s} \sim N(\lambda, 1), \qquad \lambda = \frac{\tau}{s}.$$

The quantity $\lambda$ is the standardized signal relative to its estimation noise. Under $H_0$, $\lambda = 0$. A two-sided test rejects if $|Z| > c$, where $c = z_{1-\alpha/2}$ and $z_q = \Phi^{-1}(q)$. Under a true effect $\tau = \delta$,

$$\begin{aligned} \text{Power}(\delta) &= \Pr_\delta(Z > c) + \Pr_\delta(Z < -c) \\ &= 1 - \Phi\left(c - \frac{\delta}{s}\right) + \Phi\left(-c - \frac{\delta}{s}\right). \end{aligned}$$

For positive $\delta$, the first term is the chance of detecting a positive effect; the last term is the small chance of rejecting in the opposite direction. For $\delta = 0$, this expression equals $\alpha$. As $|\delta|/s$ increases, power increases. At fixed sample size and a true effect arbitrarily close to zero, power approaches $\alpha$.

**One-sided tests.** If a positive-effect alternative is justified and specified before seeing the data, reject when $Z > z_{1-\alpha}$. Its power at $\delta > 0$ is

$$1 - \Phi\left(z_{1-\alpha} - \frac{\delta}{s}\right).$$

This gains power in the chosen direction, but does not detect effects in the opposite direction. Choosing the direction after observing the result invalidates the nominal error control.

### 4.4.3 Minimum detectable effect and sample-size planning

For a prespecified alternative $\delta_* > 0$, level $\alpha$, and target power $1 - \beta_*$, the normal approximation yields

$$\frac{\delta_*}{s} \gtrsim z_{1-\alpha/2} + z_{1-\beta_*},$$

where the opposite-tail rejection probability is neglected. With total sample size $N$, treatment fraction $q$, and equal arm variances $\sigma^2$, the corresponding minimum detectable effect is

$$\text{MDE} \approx (z_{1-\alpha/2} + z_{1-\beta_*})\frac{\sigma}{\sqrt{Nq(1-q)}}.$$

The MDE is the alternative at which the design attains the specified power, not a lower bound on effects that can produce a significant result. Section 4.5 derives enrollment formulas for continuous and binary outcomes and provides numerical and computational examples.

**Interpretation of planned power** A planned 80% power means an approximately 80% rejection probability over repeated experiments under the specified effect and model. It does not guarantee significance in this particular experiment. A sample-size claim also depends on valid variance estimates, independent assignment units, retention, and the assumed effect. Use historical data or pilot estimates, check plausible variance and effect scenarios, and allow for attrition. For simple independent MCAR loss with retention probability $r$, an approximate enrollment target is $N_{\text{required}}/r$. This adjustment alone does not address treatment-dependent missingness or repeated-measure requirements.

### 4.4.4 Design strategies for improving power

1. **Increase independent information.** Recruit more randomized users or clusters, or extend follow-up when that adds useful information. Repeated sessions from the same user are not independent users. In the equal-cluster illustration of Section 4.3, the design effect $1+(m-1)\rho$ inflates variance. Power calculations must respect the assignment unit, cluster sizes, and within-cluster correlation.
2. **Reduce outcome noise.** Improve measurement and use informative pre-treatment covariates through CUPED, prespecified regression adjustment, blocking, or paired designs with valid inference. In the ideal CUPED case, $\text{Var}(Y_{\text{adj}}) = \text{Var}(Y)(1 - \rho_{YX}^2)$. Under comparable

arm variance reductions, $\rho_{YX} = 0.6$ reduces the required sample size to roughly 64% of the unadjusted target. Adjustment for treatment-induced mediators changes the target and is not a routine variance-reduction device.

3. **Choose allocation deliberately.** For equal arm variances and a fixed total number of equally costly independent units, $q = 1/2$ minimizes variance because it maximizes $q(1-q)$. With unequal variances, minimizing $\sigma_1^2/q + \sigma_0^2/(1-q)$ yields

$$q^* = \frac{\sigma_1}{\sigma_1 + \sigma_0}.$$

   Risk, recruitment costs, and interference can require a different design.

4. **Choose a metric and alternative that match the question.** Prespecify a sensitive, reliable primary metric and a meaningful effect. A justified directional test can gain power. Selecting a favorable metric, subgroup, or direction after examining results needs additional error control and independent confirmation.

5. **Manage multiple testing and repeated looks.** Testing many metrics or stopping when a usual fixed-horizon $p$-value first crosses 0.05 inflates false positives. Prespecify the primary test and use an appropriate multiplicity or sequential procedure when required. Design power using that procedure's actual rejection boundary. Simply raising $\alpha$ buys power by accepting more false alarms.

### 4.4.5 Criteria for selecting a test statistic

An estimator estimates a quantity such as $\tau$; a test statistic combines data into evidence against a null. They need not be the same object. For a statistic $T$ and rejection region $\mathcal{R}$,

$$\Pr_{H_0}(T \in \mathcal{R}) \le \alpha, \qquad \text{Power}(\theta) = \Pr_{\theta}(T \in \mathcal{R}).$$

The important question is not whether $T$ has a large numerical value, but whether its distribution under the relevant alternative separates from its correctly calibrated null distribution.

**Null calibration.** A statistic needs a valid null reference distribution, including treatment assignment, clustering, nuisance estimation, and any selection steps. For example, the standardized statistic

$$T = \frac{\widehat{\tau}}{\widehat{\text{SE}}(\widehat{\tau})}$$

compares an estimate with its uncertainty. A large estimate can be weak evidence when its standard error is also large. Incorrectly treating correlated rows as independent makes the denominator too small and can create apparently high detection rates by inflating Type I error.

**Sensitivity to the intended alternative.** At a fixed valid level, prefer statistics with high power against effects that matter scientifically. Squaring or taking an absolute value targets both signs; a signed statistic targets a direction. A mean-based statistic can miss heterogeneous effects that cancel in the mean, whereas a statistic of variability or predictive structure asks a different question. In the simple-versus-simple case, the Neyman–Pearson principle chooses a large likelihood ratio

$$\Lambda(D) = \frac{f_1(D)}{f_0(D)}$$

as evidence against $H_0$. This maximizes power at a fixed level for the specified pair of distributions; it does not identify one universally best statistic for every composite alternative.

**Efficiency, robustness, and reproducibility.** A useful statistic captures signal with limited noise, works under the assumptions plausible for the experiment, and remains computable under sparse exposures, ties, outliers, and small samples. Lower estimator variance helps power when the estimand and bias are comparable. Mean squared error $\mathrm{MSE}(\widehat{\tau}) = \mathrm{Var}(\widehat{\tau}) + \mathrm{Bias}(\widehat{\tau})^2$ alone does not establish test validity. A statistic should also be prespecified or have selection accounted for, with a reproducible calculation applied consistently to observed and resampled data. Multiplying $T$ by a positive constant does not improve a rank-based randomization test: it preserves every ordering and its $p$-value.

### 4.4.6 Connection to interference randomization tests

A conditional randomization test for interference can hold focal units' own treatments fixed while resampling auxiliary units' assignments from the conditional assignment distribution. Under the no-interference null, focal outcomes are invariant to these resampled assignments. Exposures and the test statistic are recomputed using the same rule for each draw. A possible statistic is $T = |\widehat{\gamma}_H|$, where $\widehat{\gamma}_H$ is the exposure coefficient in a focal-unit regression of $Y$ on $A$, $X$, and $H$. With exchangeable observed and resampled statistics, the Monte Carlo value

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T^{(b)} \geq T^{(0)}\}}{B + 1}$$

is super-uniform under the null, so its rejection probability is at most $\alpha$. Appropriate exposure definitions and sensitive statistics improve power; validity follows from the randomization argument, not from a claim that this regression is the true outcome model. Any data-adaptive fitting or selection must preserve that argument, for example by repeating the same procedure for every resampled dataset.

Using an uninformative exposure can yield a valid but low-power test. Increasing $B$ reduces Monte Carlo variability and improves $p$-value resolution; it does not create more experimental information. Since the smallest possible $p$-value is $1/(B+1)$, rejection at $p \leq 0.05$ requires at least $B = 19$, but substantially more draws are usually needed for stable inference near the cutoff. If $T$ is constant across all draws, the above formula gives $p = 1$: perfect Type I error control with no detection power.

**Design implications.** Power is defined relative to a specified alternative and a calibrated rejection rule. Increasing independent information and reducing residual variance can improve power without increasing the Type I error rate. Prespecification of the estimand, metric, sample size, and decision rule links these design choices to interpretable error probabilities.

## 4.5 Sample Size Determination

Sample-size determination specifies the number of independent randomization units required to attain a target power at a prespecified effect size. The calculations below specialize the preceding power results to binary and continuous outcomes under two independent arms, a fixed final analysis, and a two-sided test.

### 4.5.1 Planning parameters

Specify the primary outcome and its measurement window, the randomization unit, the target population, and the treatment comparison. Then choose:

1. The baseline conversion probability $p_0$ for a binary outcome, or arm-specific variances $\sigma_0^2, \sigma_1^2$ for a continuous outcome, estimated from comparable historical data or a pilot.
2. A minimum detectable effect (MDE) $\Delta_* > 0$, representing a scientifically or commercially meaningful planning alternative.
3. The Type I error level $\alpha$, commonly 0.05, and power $1 - \beta$, commonly 0.80 or 0.90.
4. The allocation ratio $r = n_1/n_0$, and whether the test is one-sided or two-sided. Fix the direction of a one-sided test in advance.

Here $n_a$ is the number of independent units in arm $a$, and $N = n_0 + n_1$ is the total. An MDE specifies an effect at which the design should attain the target power; it is not a guarantee of significance or a lower bound on significant estimates.

### 4.5.2 Binary outcomes: planning for a conversion-rate difference

For equal allocation, let each arm contain $n$ independent users and write

$$p_a = \Pr(Y(a) = 1), \qquad \Delta_* = p_1 - p_0 > 0, \qquad \bar{p} = (p_0 + p_1)/2.$$

The difference in sample conversion rates has alternative variance

$$\mathrm{Var}(\widehat{p}_1 - \widehat{p}_0) = \frac{p_1(1-p_1) + p_0(1-p_0)}{n}.$$

For the usual pooled two-proportion test, a common normal planning approximation uses a null critical threshold $z_{1-\alpha/2}\sqrt{2\bar{p}(1-\bar{p})/n}$. Define

$$v_0 = 2\bar{p}(1-\bar{p}), \qquad v_1 = p_0(1-p_0) + p_1(1-p_1).$$

Ignoring the small probability of rejection in the opposite direction, the power is approximately

$$\Pr\left(\widehat{p}_1 - \widehat{p}_0 > z_{1-\alpha/2}\sqrt{v_0/n}\right) \approx \Phi\left(\frac{\sqrt{n}\,\Delta_* - z_{1-\alpha/2}\sqrt{v_0}}{\sqrt{v_1}}\right).$$

Setting this expression to $1 - \beta$ and solving for $n$ gives

$$\boxed{n \approx \frac{\left[z_{1-\alpha/2}\sqrt{2\bar{p}(1-\bar{p})} + z_{1-\beta}\sqrt{p_0(1-p_0) + p_1(1-p_1)}\right]^2}{\Delta_*^2}.}$$

Round up the result for *each* arm. This is an asymptotic approximation without a continuity correction; check actual power using a procedure matched to the planned test when event counts are small or conversion is very rare. The earlier baseline-only approximation replaces both arm variances by $p_0(1-p_0)$ and can give a different enrollment target.

**Example: 10% to 11% conversion.** For $p_0 = 0.10$, $p_1 = 0.11$, $\alpha = 0.05$, and power 0.80,

$$\Delta_* = 0.01, \qquad n \approx 14{,}750.79, \qquad n_0 = n_1 = 14{,}751, \quad N = 29{,}502.$$

The effect is one *percentage point*, equivalent to a 10% relative increase: $(0.11 - 0.10)/0.10 = 0.10$. Distinguishing these two scales is essential when translating a business requirement into an MDE.

### 4.5.3 Continuous outcomes and unequal allocation

For an outcome such as revenue or viewing time, let $\Delta_* = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] > 0$. Under independent sampling,

$$\mathrm{Var}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

With allocation $n_1 = rn_0$, the usual normal approximation gives

$$n_0 \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2(\sigma_0^2 + \sigma_1^2/r)}{\Delta_*^2}, \qquad n_1 \approx rn_0.$$

For equal variances $\sigma_0^2 = \sigma_1^2 = \sigma^2$ and equal allocation,

$$\boxed{n \approx \frac{2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2}{\Delta_*^2}.}$$

Equal allocation minimizes total sample size when variances and per-unit costs are equal. Holding other inputs fixed, halving the MDE requires approximately four times as many units. For heavy-tailed revenue, ratio metrics, or complicated estimators, simulate the complete planned analysis under plausible alternatives using representative historical data.

### 4.5.4 Python implementation

The following standard-library code implements the binary planning formula directly, so its output matches the numerical example above.

```
from math import ceil, sqrt
from statistics import NormalDist

p0, p1 = 0.10, 0.11
alpha, power = 0.05, 0.80
normal = NormalDist()
z_alpha = normal.inv_cdf(1 - alpha / 2)
z_power = normal.inv_cdf(power)
pbar = (p0 + p1) / 2

n = ceil(
    (z_alpha * sqrt(2 * pbar * (1 - pbar))
     + z_power * sqrt(p0 * (1 - p0) + p1 * (1 - p1)))**2
    / (p1 - p0)**2
)
print("Per arm:", n)       # 14751
print("Total:", 2 * n)     # 29502
```

#### 4.5.5 Extensions to complex experimental designs

- **Randomization units and clustering.** Count users when users are randomized, rather than treating sessions as independent users. Cluster-randomized designs require cluster-level planning, accounting for within-cluster correlation and the number and sizes of clusters.
- **Multiple comparisons and monitoring.** Design for the prespecified family of primary comparisons. For $K$ comparisons, a simple Bonferroni plan replaces $\alpha$ by $\alpha/K$. Repeated significance checks or early stopping require a sequential design with its own power calculation; the fixed-horizon formula does not justify arbitrary peeking.
- **Precision adjustment.** With a stable CUPED variance reduction factor $1-R^2$, the required sample size is approximately multiplied by $1-R^2$ for the same estimand and testing procedure. Estimate the achievable reduction from representative data rather than assuming it.
- **Retention and experiment duration.** Under independent MCAR loss with retention rate $r_{\text{ret}}$, enroll approximately $N/r_{\text{ret}}$ units. This does not repair treatment-dependent missingness. Divide enrollment by the daily rate of newly eligible, unique randomized units to estimate recruitment time, then allow completion of the outcome window and relevant calendar cycles.
- **ATE versus targeting.** A design powered for an overall ATE need not be powered for CATE, a treatment–covariate interaction, or a top-20% policy. Plan for independent policy evaluation, subgroup event counts, and training/validation/test allocation; use simulation when the policy-learning procedure is part of the target analysis.

An analysis plan should record the baseline assumptions, absolute and relative MDE, test, allocation, power, enrollment target, and stopping rule. Repeat the planning calculation over plausible baseline rates or variances to assess how sensitive the target is to uncertainty in historical inputs.